

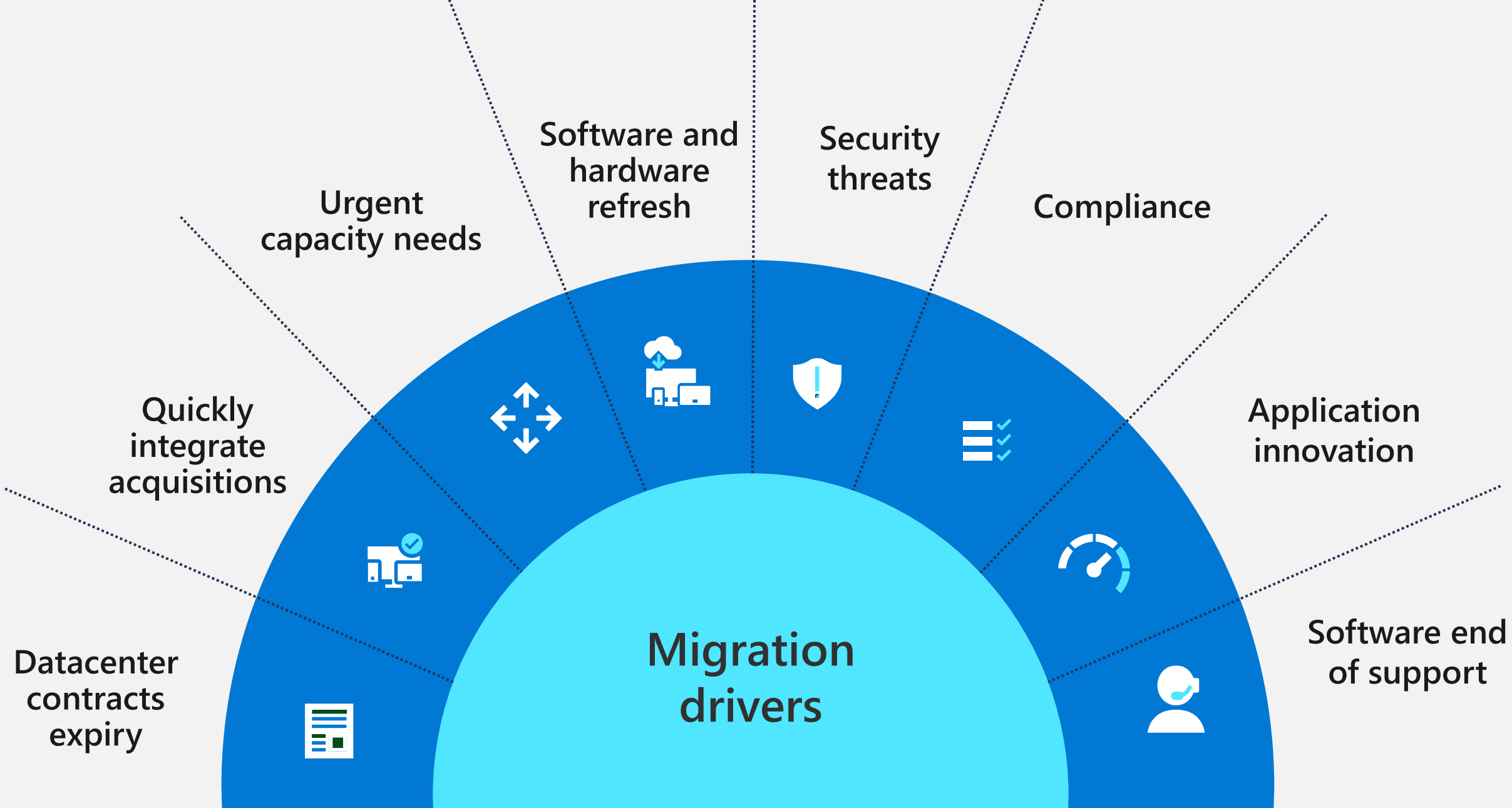
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Microsoft Azure Virtual Training Day: Modernize .NET Apps

Assessing your Migration



Migration Journey

 Assess	 Migrate	 Optimize	 Secure & Manage
- Create Cloud Migration Plan - Involve Stakeholders - Calculate your TCO - Discover and Evaluate Apps	Define Strategy: - Refactor - Rearchitect - Rebuild Choose the appropriate tools for migration: - Azure Migrate - Azure Site Recovery	- Manage and monitor costs with Azure Cost Management - Save with Azure Offers (Azure Hybrid Benefit and Azure Reservations) - Re-invest and do more with capacity and capabilities of the cloud	- Industry leading protection with Security Center - Protect your data in the cloud with Azure Backup, Azure Site Recovery,.. - Monitor your cloud health with Azure Monitor

In this session...



Preparing for the migration



Using Azure Migrate



Assessing the Web app with the App Service Migration Assistant



Assessing the database with the Data Migration Assistant

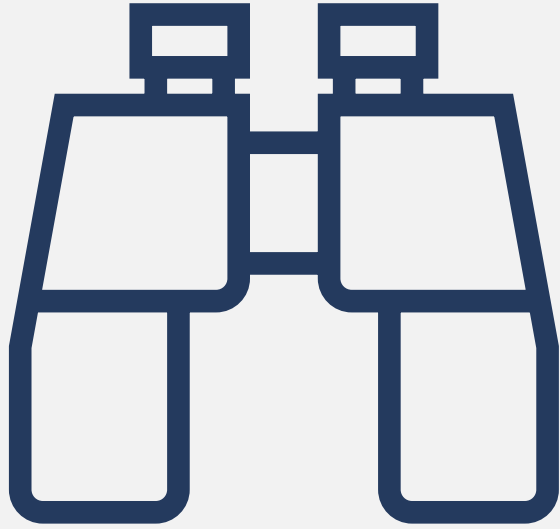


.NET to .NET Core assessment with the Portability Analyzer



Before you even migrate...

Discovery & Assessment Phase



Key to success

Discovery dependencies

Staff Training

End Users

Documentation

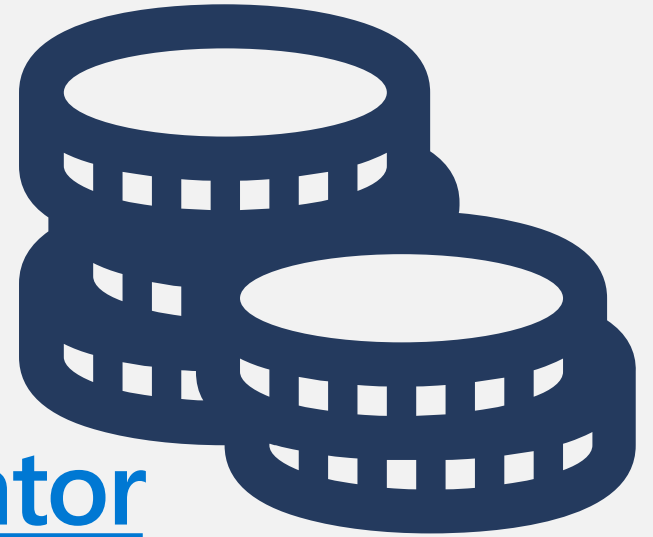
Cost

How much does it currently cost?

What would it cost in the cloud?

<https://aka.ms/mod10/tco>

<https://aka.ms/mod10/pricing-calculator>



Migrate Phase

1



Don't tackle your most important workload first

2



Communication

3



Management, monitoring, backup, disaster recovery

4



Design choices

Optimize Stage



This will become a BAU (Business As Usual) activity.

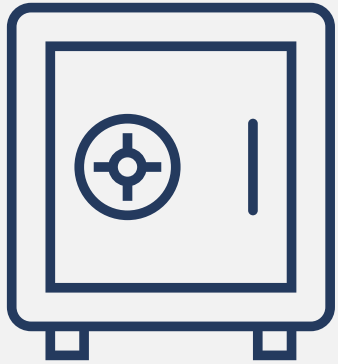


Start to track and understand your cost trends.



Work on making workloads operational better.

Secure and Manage Phase



This will become a BAU activity.

Know your responsibilities.

Staff Training

Azure Migration Center

Guidance

Migration journey
and strategies

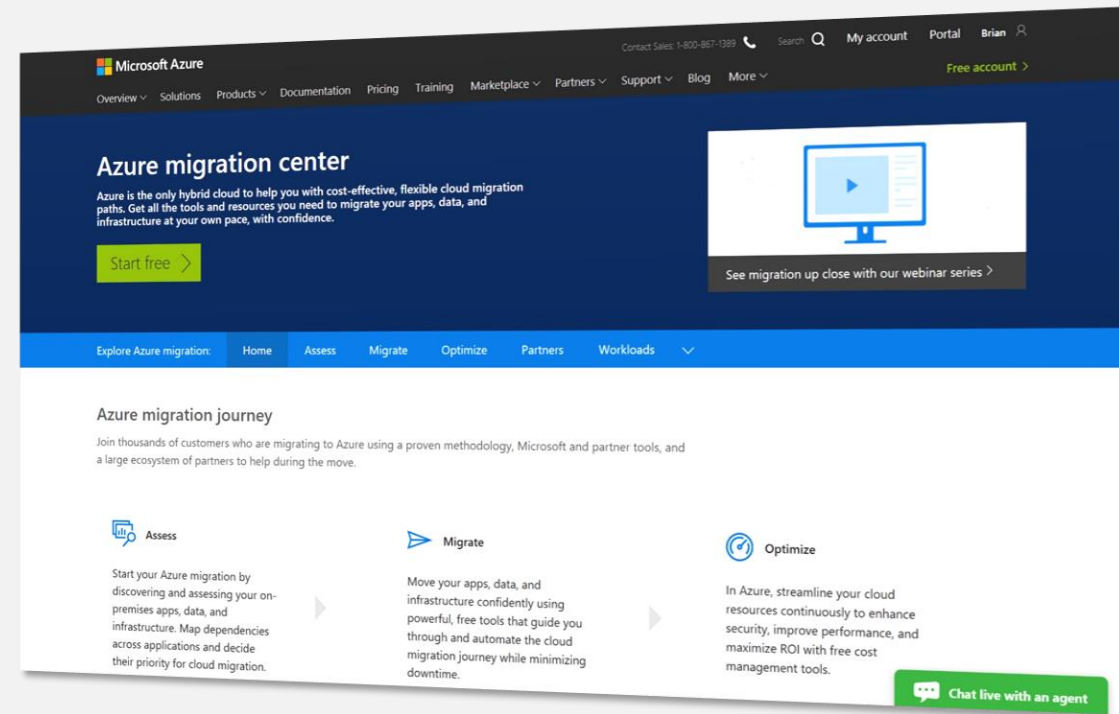
Tools

Right tools for the job

Partners

Migration experts

<https://aka.ms/mod10/migration>



DEMO

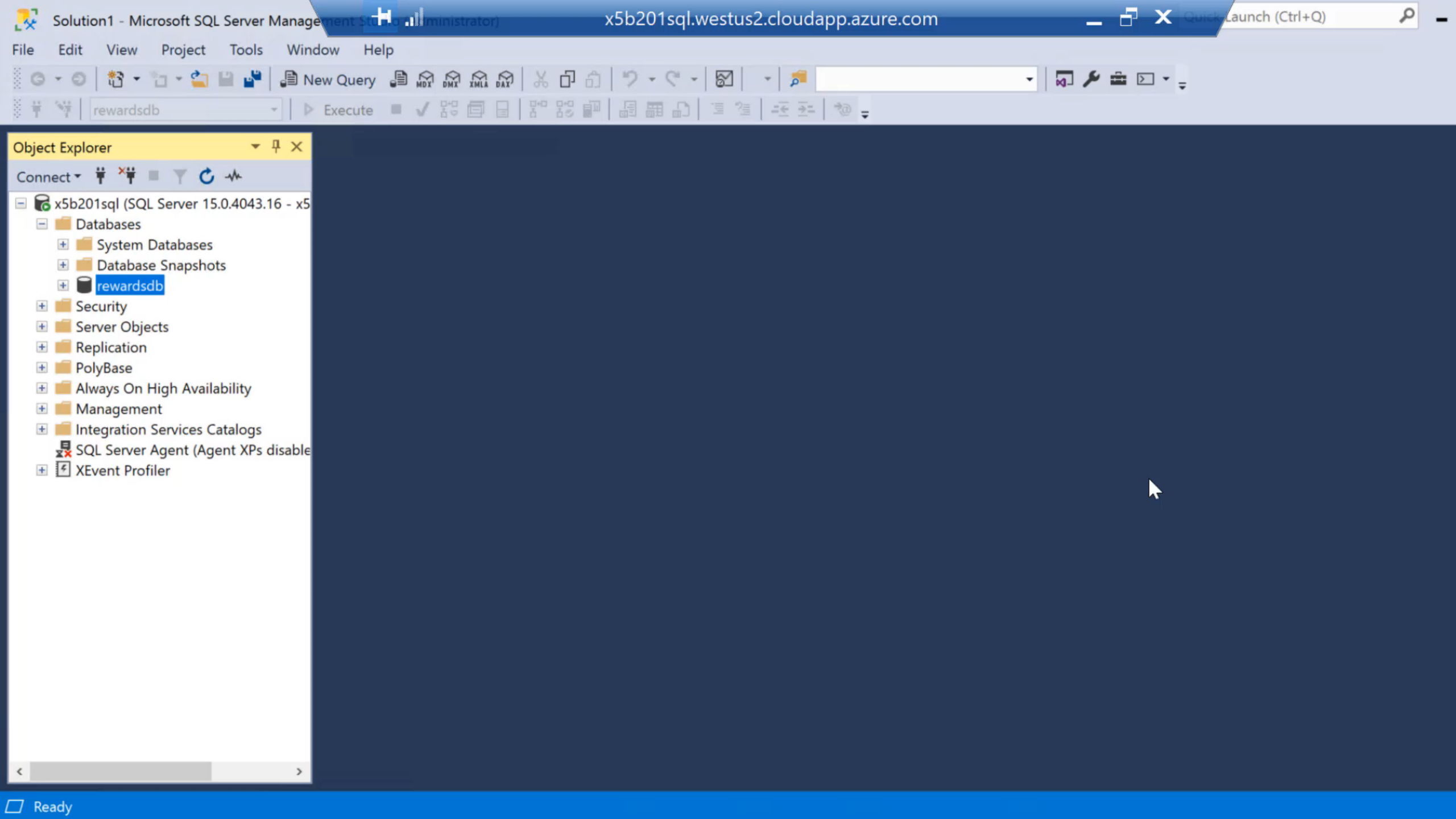
Assessing your migrations





Azure Active Directory

23 hours ago





Upgrading .NET

.NET on Azure

.NET Classic

- ASP.NET 3.5/4.x
- .NET Framework
- Runs on Windows only
- Good for existing apps

.NET Core

- .NET Core
- .NET 5
- Runs on Windows and Linux
- Great for new apps or migration
- Performance enhancements

.NET to .NET Core migration path



Assess the migration with the Portability Analyzer



Put your business logic code in .NET Standard libs



Enjoy the performance improvements!



DEMO

Assessing .NET to .NET Core

File

Edit

View

Project

Build

Debug

Test

Analyze

Tools

Extensions

Window

Help

Search (Ctrl+Q)

TailwindTrade...wards.Website

Live Share

Debug

Any CPU

IIS Express (Microsoft Edge)

Toolbox

Search Toolbox

General

There are no usable controls in this group. Drag an item onto this text to add it to the toolbox.

TailwindTraders.Rewards.Website

Login.aspx

Application

Build

Web

Package/Publish Web

Package/Publish SQL

Build Events

Resources

Settings

Reference Paths

Signing

Code Analysis

Configuration: N/A

Platform: N/A

Assembly name: Tailwind.Traders.Rewards.Web

Default namespace: Tailwind.Traders.Rewards.Web

Target framework: .NET Framework 3.5

Output type: Class Library

☐ Auto-generate binding redirects

Startup object: (Not set)

Resources

Specify how application resources will be managed:

☒ Icon and manifest

A manifest determines specific settings for an application. To embed a custom manifest, first add it your project and then select it from the list below.

Icon: (Default Icon)

Manifest: Embed manifest with default settings

☐ Resource file:

Solution Explorer

Search Solution Explorer (Ctrl+;)

Solution 'TailwindTraders.Rewards.V

TailwindTraders.Rewards.Web

Connected Services

Properties

References

Content

Data_Layer

Models

Admin.aspx

Admin.Master

Create.aspx

Default.aspx

favicon.ico

Login.aspx

packages.config

Site.Master

Update.aspx

Web.config

Output

Ready

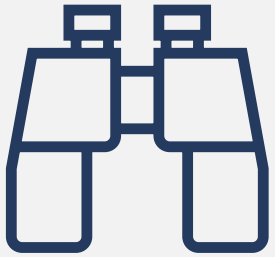
0

7

TailwindTraders-Rewards

Detached at 0b1575e8

In review...



In review...



Preparing for the migration



Using Azure Migrate



Assessing the Web app with the App Service Migration Assistant



Assessing the database with the Data Migration Assistant



.NET to .NET Core assessment with the Portability Analyzer

Call to action!

Explore overviews,
tutorials, samples, and more.

aka.ms/mod10

Home / Azure migration center

Azure migration center

Let's go forward, together. Save money and improve efficiency by migrating existing workloads to Azure.

[Get expert help >](#)

[Start your migration with an Azure free account >](#)

Home Migration journey > Migration program > Resources > Migration partners > Webinars > Workloads v

Migrate efficiently on your own terms



Optimize costs and migrate data with confidence

Save money with the [most cost-effective offers](#) for Windows Server and SQL Server. Confidently migrate your workloads to Azure with [best practices](#), [expert guidance](#), and [cost optimization tools](#).

[Learn more >](#)



Stay secure and resilient across hybrid environments

Protect workloads across your hybrid environments with [intelligent security services](#) backed by 3,500 cybersecurity professionals. Use [built-in resiliency](#) to avoid costly business interruptions.



Scale your workloads and applications on demand

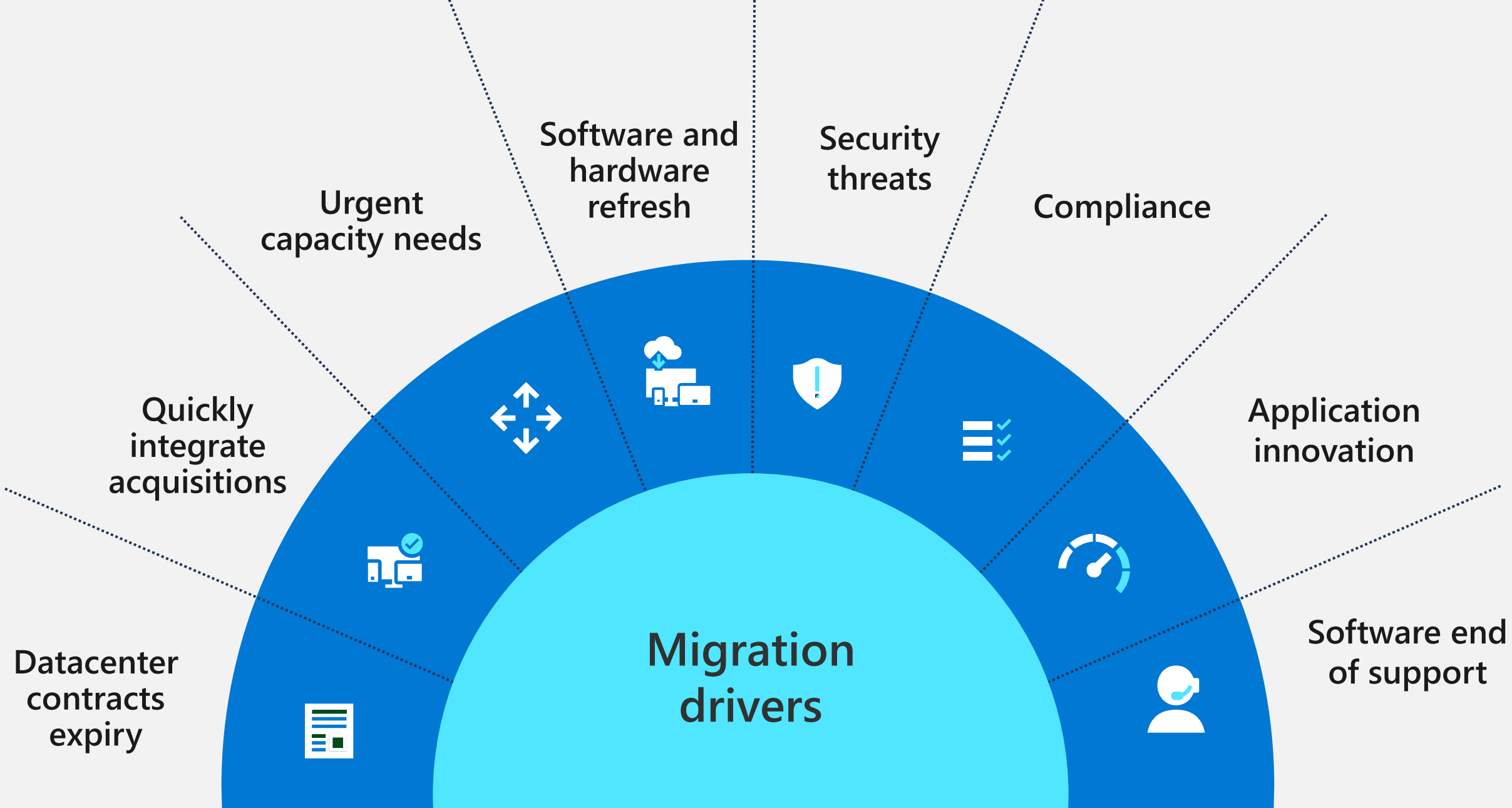
Increase agility with best-in-class [Azure infrastructure](#) that scales to your business needs. Reduce operational burden with fully managed [application](#) and [database](#) services in Azure.

[Hear from customers >](#)

Evaluate your migration readiness with the [Strategic Migration Assessment and Readiness Tool](#) >



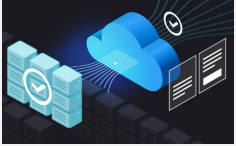
Moving Application Data to the Cloud



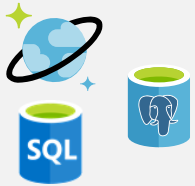
Migration Journey

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In this session...



Meeting the tools, Azure Migrate



Data solutions on Azure



Migrating the SQL database



Optimizing the services



Taking advantage of Azure Storage



Data Migration Tools

Azure Database Migration Guide

Select your source and target to get started

[Need a recommendation?](#)

Microsoft
SQL Server

ORACLE®

DB2



Microsoft Azure
Table Storage

SQL Server Migration Assistant

From:
Access, DB2,
MySQL, SAP



SQL Server Migration Assistant

From:

Access, DB2,
MySQL, SAP

To:

SQL Server, Azure
SQL DB





Data Solutions on Azure



Azure Cosmos DB

..... more coming soon

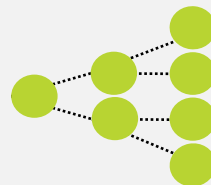
Key-Value



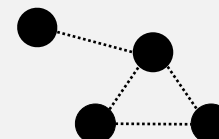
Column-family



Documents



Graph



Global distribution Elastic scale out Guaranteed low latency Five consistency models Comprehensive SLAs

A globally-distributed, multi-model database service

Azure Database for open source DBs engines



Fully managed services
without changing your
applications

Simple and flexible
pricing, Language and
frameworks of your choice

Automatic backups,
patching, built-in
monitoring, security



Key takeaways

Fully managed PaaS

99.99% availability guaranteed

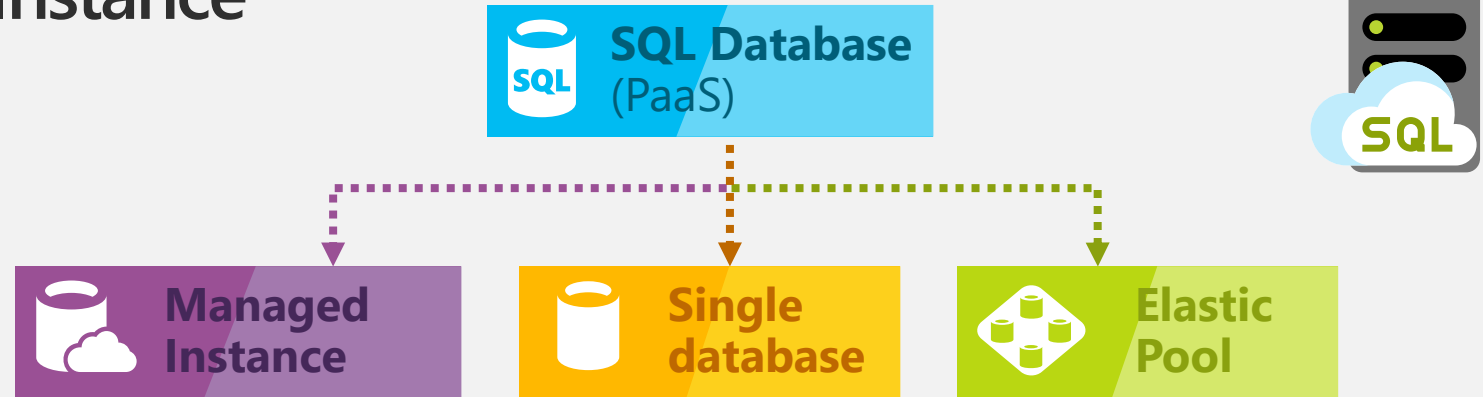
Based on well known SQL Server technology

Single database / Elastic pool

Full monitoring and alerting capacity

Azure SQL Database Managed Instance

New deployment option enabling friction-free migration of SQL Server workloads to a fully-managed service



Fully-managed service

- Built on the same infrastructure as SQL Database
- Provides the same benefits (PaaS)

SQL Server compatibility

- Fully-fledged SQL instance with nearly 100% compat with on-premise

Full isolation and security

- Contained within your VNet
- Private IP addresses
- Express Route / VPN connectivity

New purchasing options

- Transparent
- Frictionless
- Competitive

DEMO



Creating the Azure SQL Database

 Rewards

Resource group

 Search (Ctrl+ /)

Overview

Activity log

Access control (IAM)

Tags

Events

Settings

Quickstart

Deployments

Policies

Properties

Locks

Export template

Cost Management

 Add

 Edit columns

 Delete resource group

 Refresh

 Move

 Export to CSV



Essentials

 Filter by name...

Type == **all** 

Location == **all** 

 Add filter

Showing 1 to 4 of 4 records.

☐ Show hidden types 

No grouping 

List view

☐ Name 	Type 	Location 
☐  Rewards-Key-Vault	Key vault	West Europe
☐  tailwindrewardsonazure	App Service	West Europe
☐  tailwindrewardsonazure	Application Insights	West Europe
☐  tailwindrewardsonazure-a0c0	App Service plan	West Europe

Azure Database Migration Service



Azure
SQL DB



Azure
SQL DB MI



Azure
SQL VM



Azure
Cosmos DB



Azure DB
for MySQL



Azure DB
for PostgreSQL

SQL Server	X	X	X			
AWS SQL	X					
Oracle						X
DB2	X					
MongoDB				X		
MySQL					X	
AWS MySQL					X	
PostgreSQL						X
AWS PostgreSQL						X



DEMO

Migrating the SQL data

Rewards

Resource group

[Overview](#)[Activity log](#)[Access control \(IAM\)](#)[Tags](#)[Events](#)

Settings

[Quickstart](#)[Deployments](#)[Policies](#)[Properties](#)[Locks](#)[Export template](#)[Cost Management](#)

[+ Add](#) [≡ Edit columns](#) [🗑 Delete resource group](#) [🔄 Refresh](#) [➔ Move](#) [⬇ Export to CSV](#) [⋮](#)

Essentials

[Type == all](#) [✕](#)[Location == all](#) [✕](#)[+ Add filter](#)

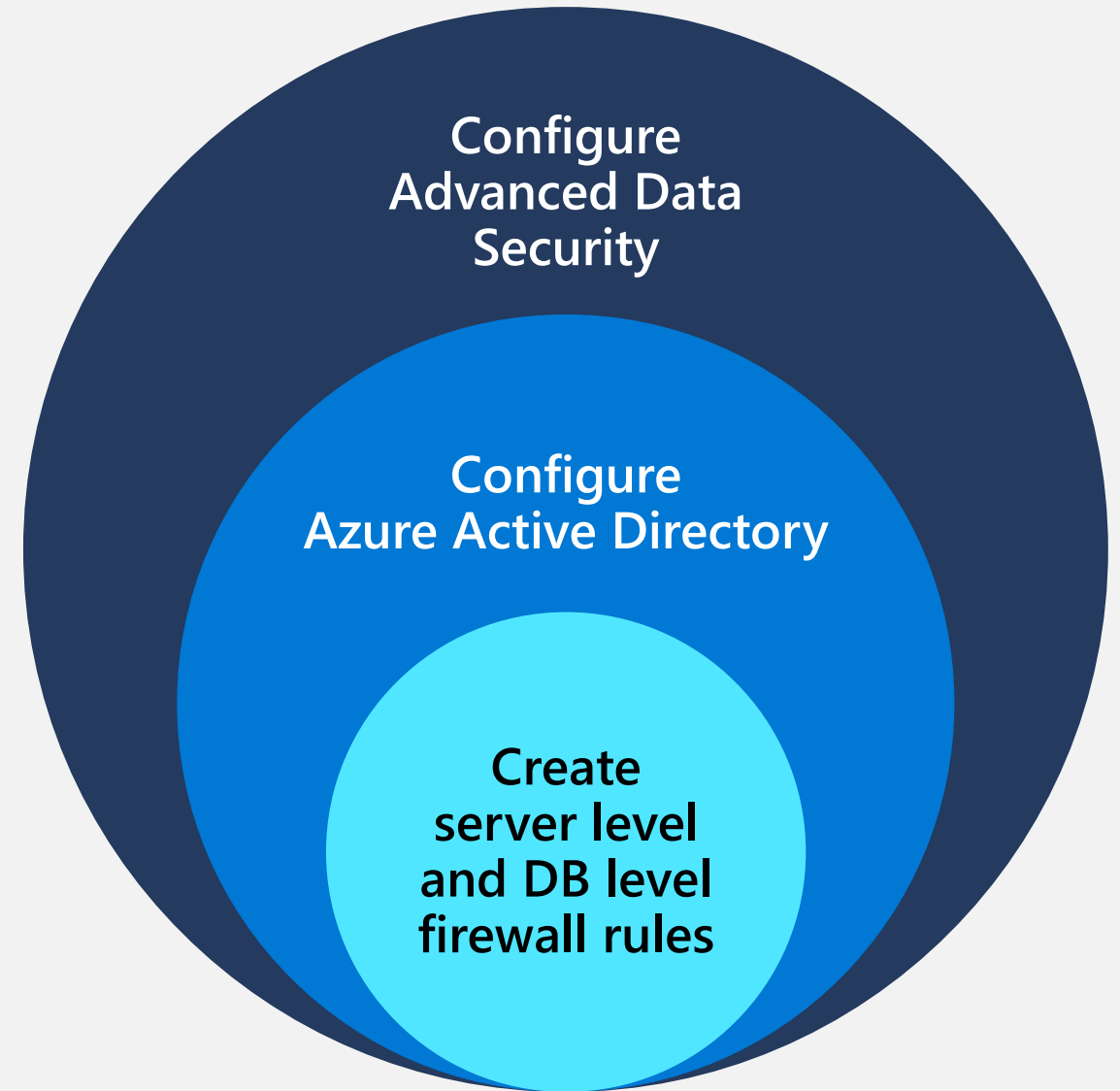
Showing 1 to 11 of 11 records.

☐ [Show hidden types](#) [ⓘ](#)[No grouping](#) [⌵](#)[List view](#)☐ [Name](#) [↑↓](#)[Type](#) [↑↓](#)[Location](#) [↑↓](#)☐ [SQL lbrewardssql](#)[SQL server](#)[West Europe](#)☐ [NIC-3r3km3qe4qrmips3xi7u4xmp](#)[Network interface](#)[West Europe](#)☐ [Key Rewards-Key-Vault](#)[Key vault](#)[West Europe](#)☐ [SQL rewardsdb \(lbrewardssql/rewardsdb\)](#)[SQL database](#)[West Europe](#)☐ [RewardsDms](#)[Azure Database Migration ...](#)[West Europe](#)☐ [RewardsVNet](#)[Virtual network](#)[West Europe](#)☐ [cslv5r2amuydaf2m4](#)[Storage account](#)[West Europe](#)[< Previous](#)[Page](#)[1](#)[of 1](#)[Next >](#)



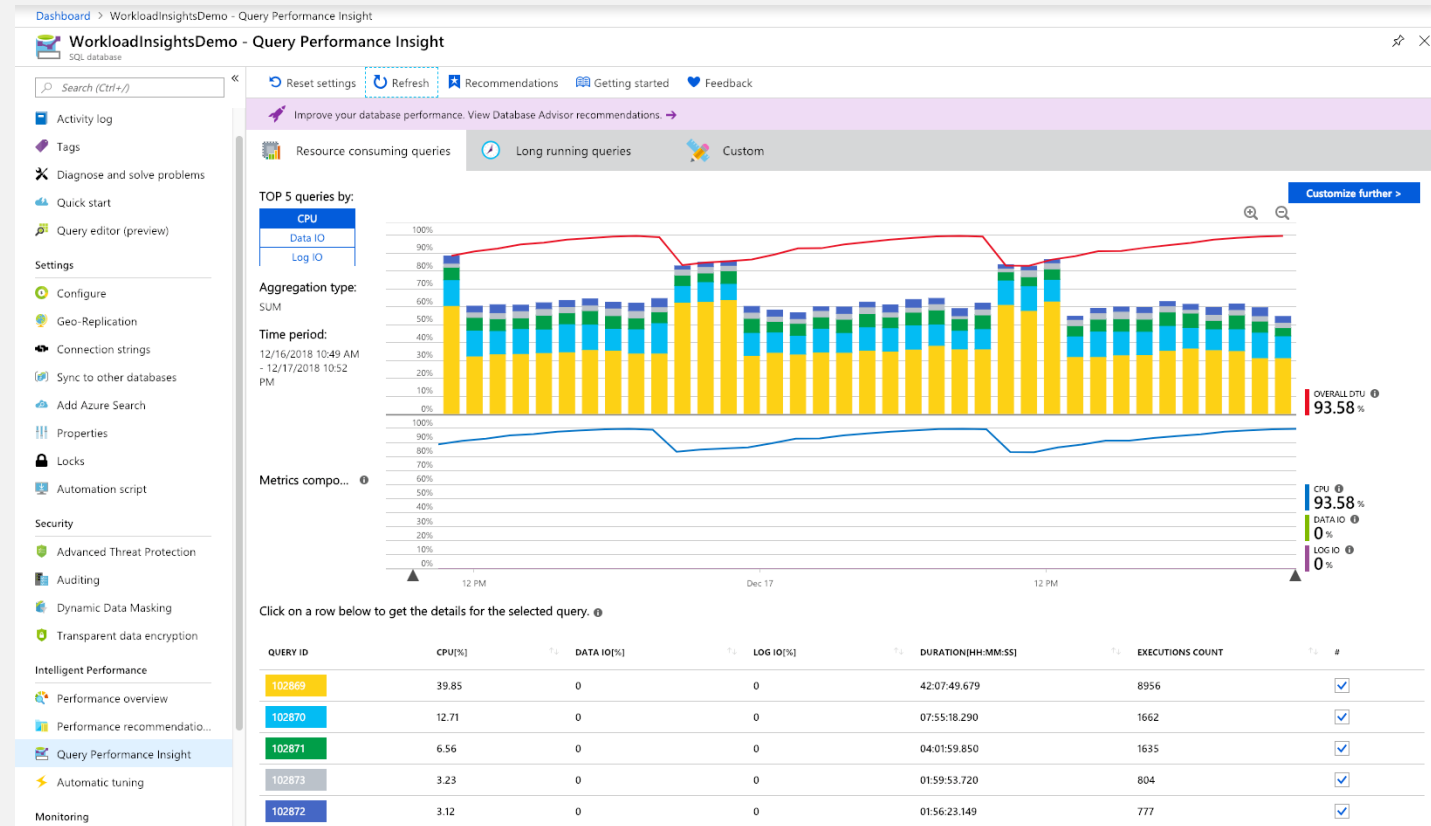
Optimizing the services

Optimizing security



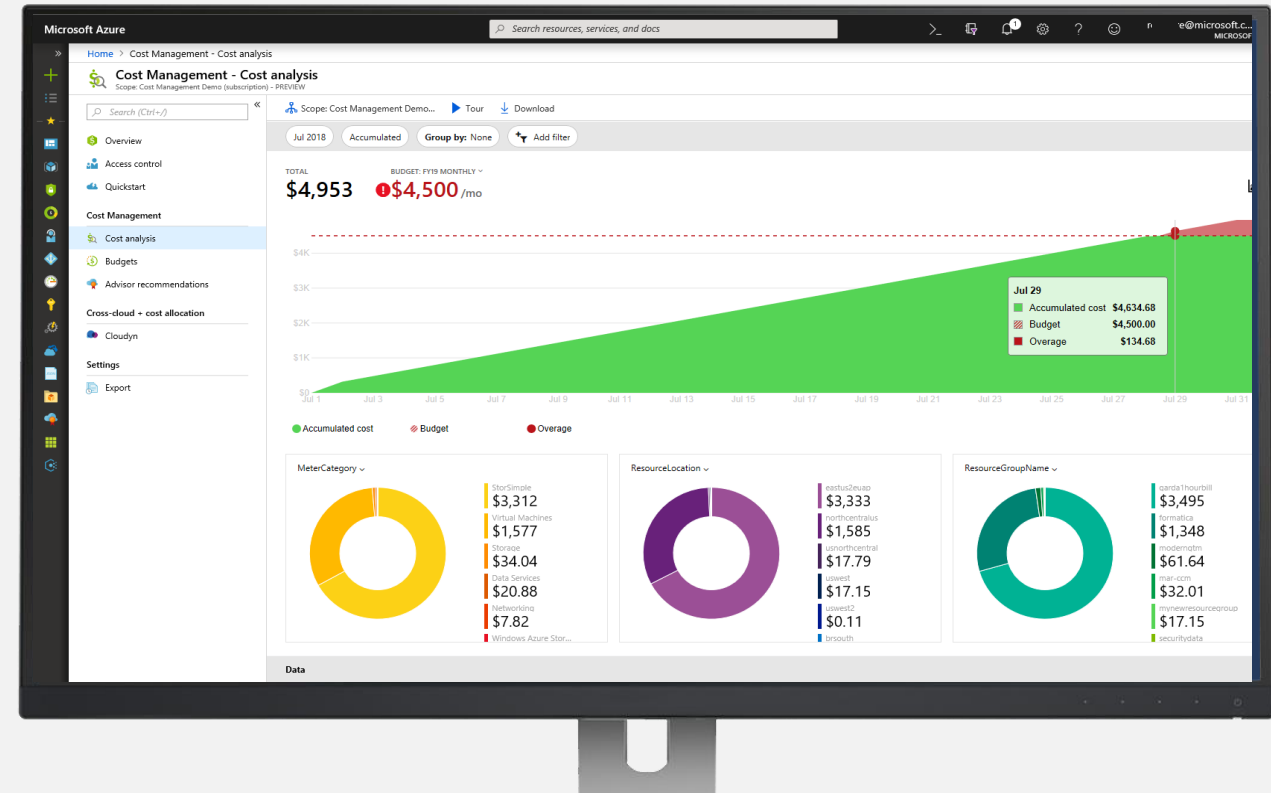
Optimizing performance of your queries

- Deep insight into DTU consumption
- CPU usage, duration, execution count
- Drill into the details
- Get recommendations



Continuously optimize migrated resources

- Azure Cost Management
- Unified experience to optimize cloud spends
- Use Azure Hybrid Benefit and Azure Reserved Instances during or after migration
- Modernize for longer term value



Optimizing for type of storage



**Structured,
relational data**
Product description



**Semi-
structured data**
Shopping cart



**Unstructured
data**
Pictures, videos

Azure Storage Services

IaaS



Storage



Virtual
machines



Networking

PaaS



Existing
frameworks



Web
and mobile



Microservices



Serverless
Compute

Disks

Persistent disks for
Azure IaaS VMs

Standard Storage Disks:
Magnetic disk based, low
IOPS, moderate latency

Premium Storage Disks:
SSD based, high IOPS,
low latency

Managed Disks

Files

Fully Managed File
Shares in the Cloud

SMB and REST access
“Lift and shift” legacy
apps

Blobs

Highly scalable, REST
based cloud object
store

Block Blobs: Sequential
I/O, Hot, Cool and
Archive Tiers

Page Blobs: Random-
write pattern data

Append Blobs

Tables

Massive auto-scaling
NoSQL store

Dynamic scaling based
on load

Scale to PBs of table data

Fast key/value lookups

Queues

Reliable queues at
scale for cloud
services

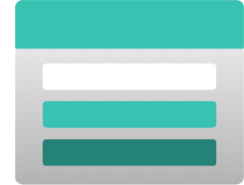
Decouple and scale
components
Message visibility
timeout and update
message to protect
against unreliable
dequeueers

Built on a Unified Distributed Storage System

Durability, Encryption at Rest, Strongly Consistent Replication, Fault Tolerance, Auto Load-Balancing

DEMO

Migrating big files to Azure Storage



**mod20lb**

App Service



Overview



Activity log



Access control (IAM)



Tags



Diagnose and solve problems



Security



Events (preview)

Deployment



Quickstart



Deployment slots



Deployment Center

Settings



Configuration



Authentication / Authorization



Browse



Stop



Swap



Restart



Delete



Get publish profile



Reset publish profile



.NET Framework 4.8 is coming to App Service starting in late July 2020 and will complete around late September 2020. Click to learn more and see progress on the deployment.

Resource group ([change](#))[lbtailwind](#)

Status

Running

Location

East US

Subscription ([change](#))[ca-lbugnion-demo-test](#)Tags ([change](#))[Click here to add tags](#)

Essentials

URL

<https://mod20lb.azurewebsites.net>

App Service Plan

[mod20lbplan \(P1v2: 2\)](#)

External Repository Project

<https://github.com/lbugnion/TailwindTraders-Website>[View as](#)

Diagnose and solve problems

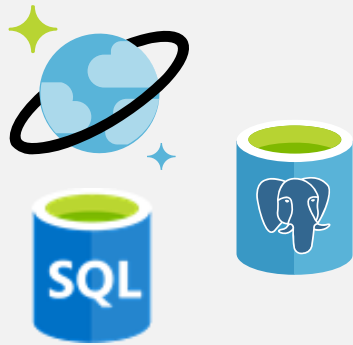
Our self-service diagnostic and troubleshooting experience helps you identify and resolve issues with your web app.



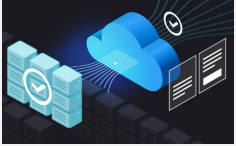
Application Insights

Application Insights helps you detect and diagnose quality issues in your apps, and helps you understand what your users actually do with it.

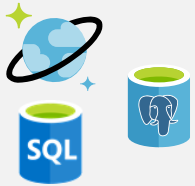
In review...



In review...



Meeting the tools, Azure Migrate



Data solutions on Azure



Migrating the SQL database



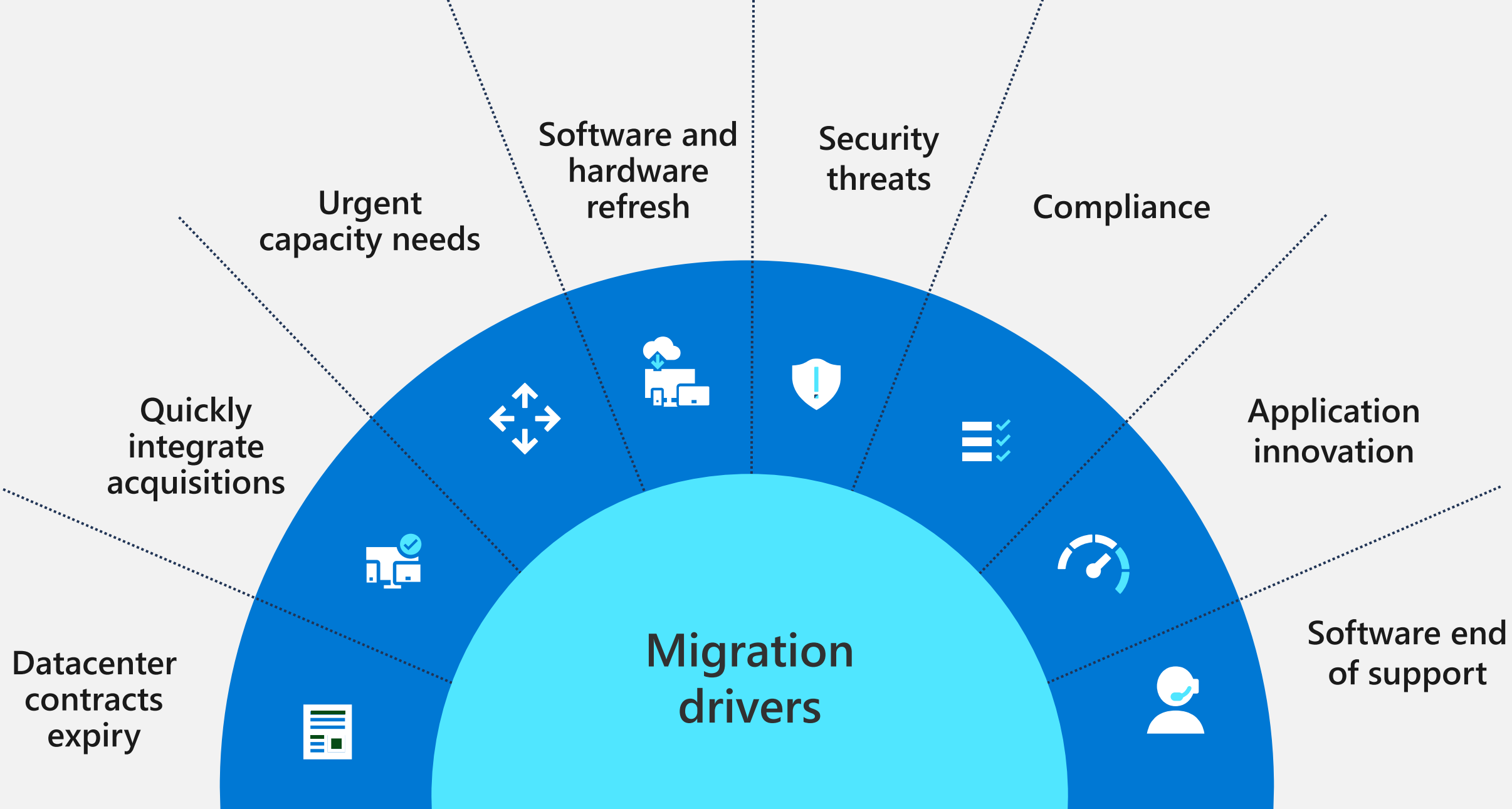
Optimizing the services



Taking advantage of Azure Storage



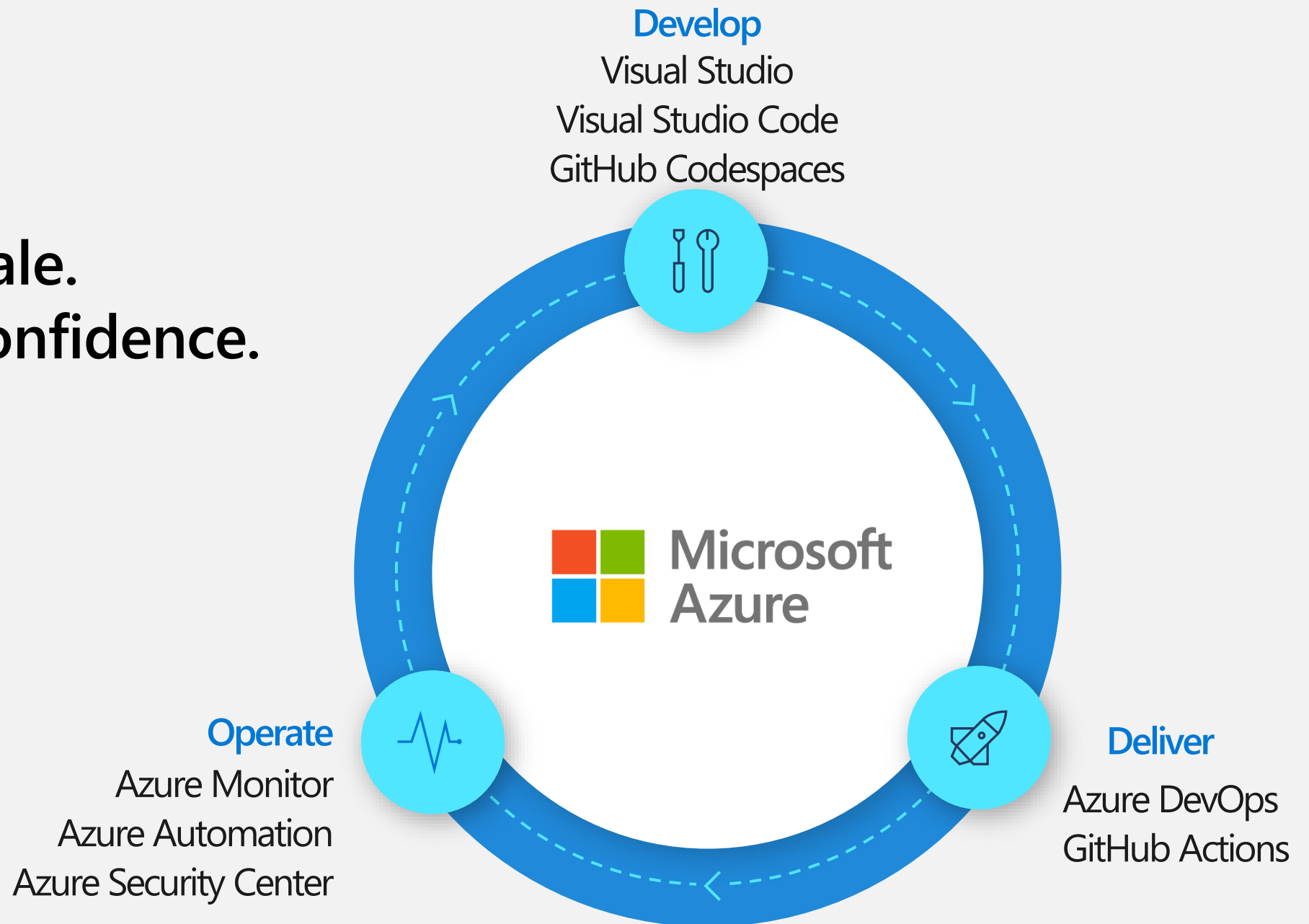
Upgrading a Web Application to App Services



Migration Journey

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**Innovate at scale.
Deliver with confidence.**



In this session...



Moving your web application to App Service



Preparing application telemetry



Managing your settings and secrets



Scaling up and Scaling out

Time To Move

How do we move from our current infrastructure to deploying in the cloud?.

Web applications hosting on Azure

1

Virtual machines

<https://aka.ms/mod20/vm-migrate>

2

Static web app on Azure Storage

<https://aka.ms/mod20/static-web-storage>

3

Azure Static Web Apps (Preview)

<https://aka.ms/mod20/static-web-apps>

4

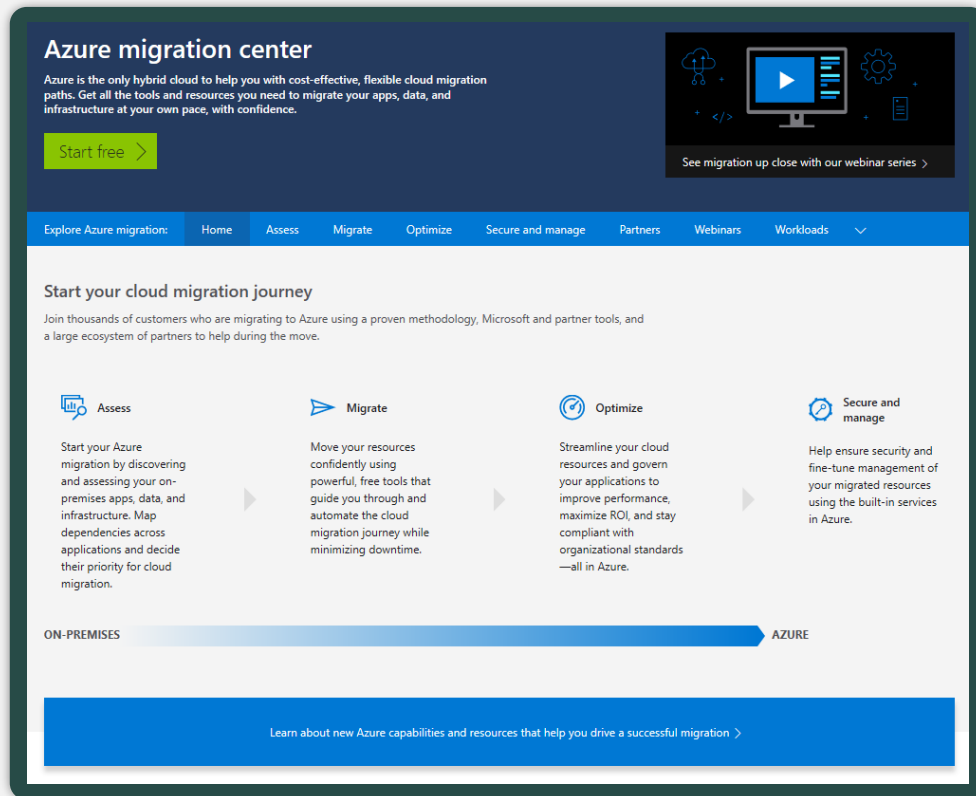
App Service with .NET, JavaScript, PHP, Java etc

<https://aka.ms/mod20/app-service>

Migration: Where to start

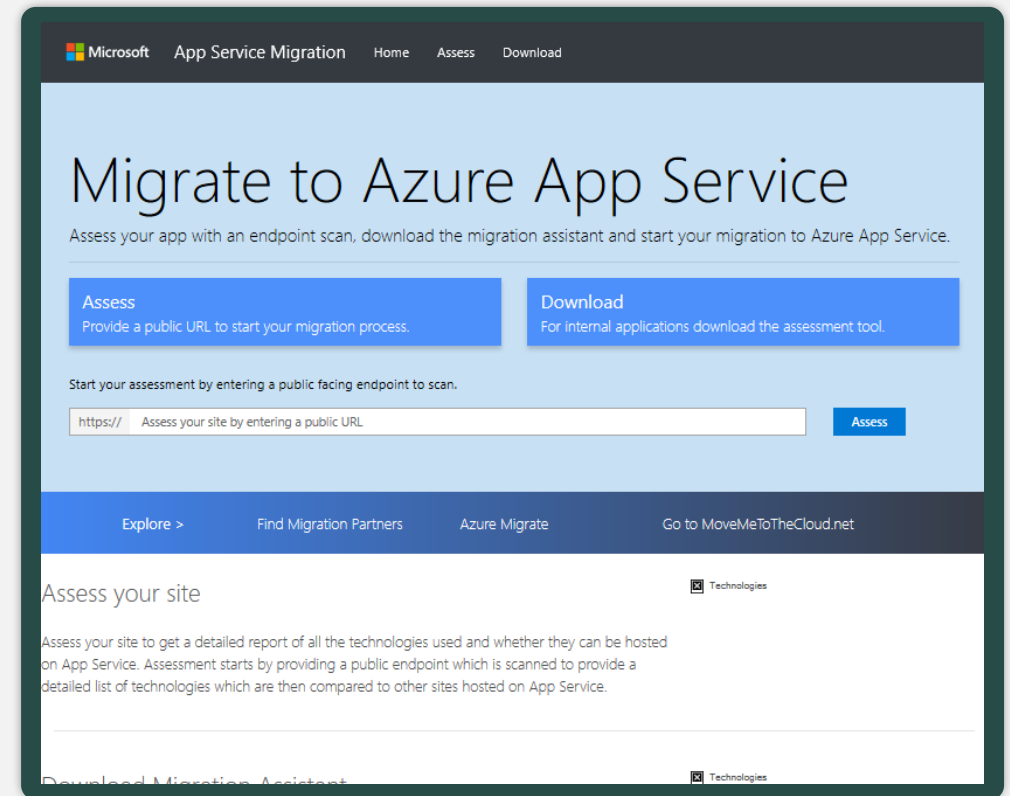
Use the Azure App Service Migration Assistant

Azure Migrate



<https://azure.microsoft.com/en-us/migration/>

Azure App Service Migration Assistant



<https://appmigration.microsoft.com/>

Azure Migrate

Choose from
over **12**
migration
tools



Azure Migrate

Choose from
over **12**
migration
tools

Assess **35K**
servers in a
single
project



Azure Migrate

Microsoft Azure

Search resources, services, and docs (G+/)

admin1@cloudmigrate...
MIGRATION SERVICES DEMO

Home > Azure migrate

Azure Migrate
Microsoft

Documentation

Search in the menu

Overview

Migration goals

Servers

Databases

Web Apps

Data

Manage

Discovered items

Migration projects

Support + Troubleshooting

New support request

What's new Get started **Tutorials & Demos**

Migrate your on-premises datacenter to Azure

With Azure migrate you can easily discover, assess and migrate your on-premises application to Azure using Microsoft or partner tools. [Learn more](#)

Windows and Linux servers

Discover, assess and migrate your on-premises VMware and Hyper-V virtual machines or Physical servers to Azure.

[Assess and migrate servers](#)

SQL and other databases

Discover, assess and migrate your on-premises databases to Azure SQL Database Managed Instance or Azure SQL Database.

[Assess and migrate databases](#)

Web apps, data and virtual desktop in...

Assess and migrate web apps, migrate data and assess virtual desktop infrastructure (VDI).

[Assess and migrate web apps](#)
[Migrate data](#)
[Assess virtual desktop infrastructure \(VDI\)](#)

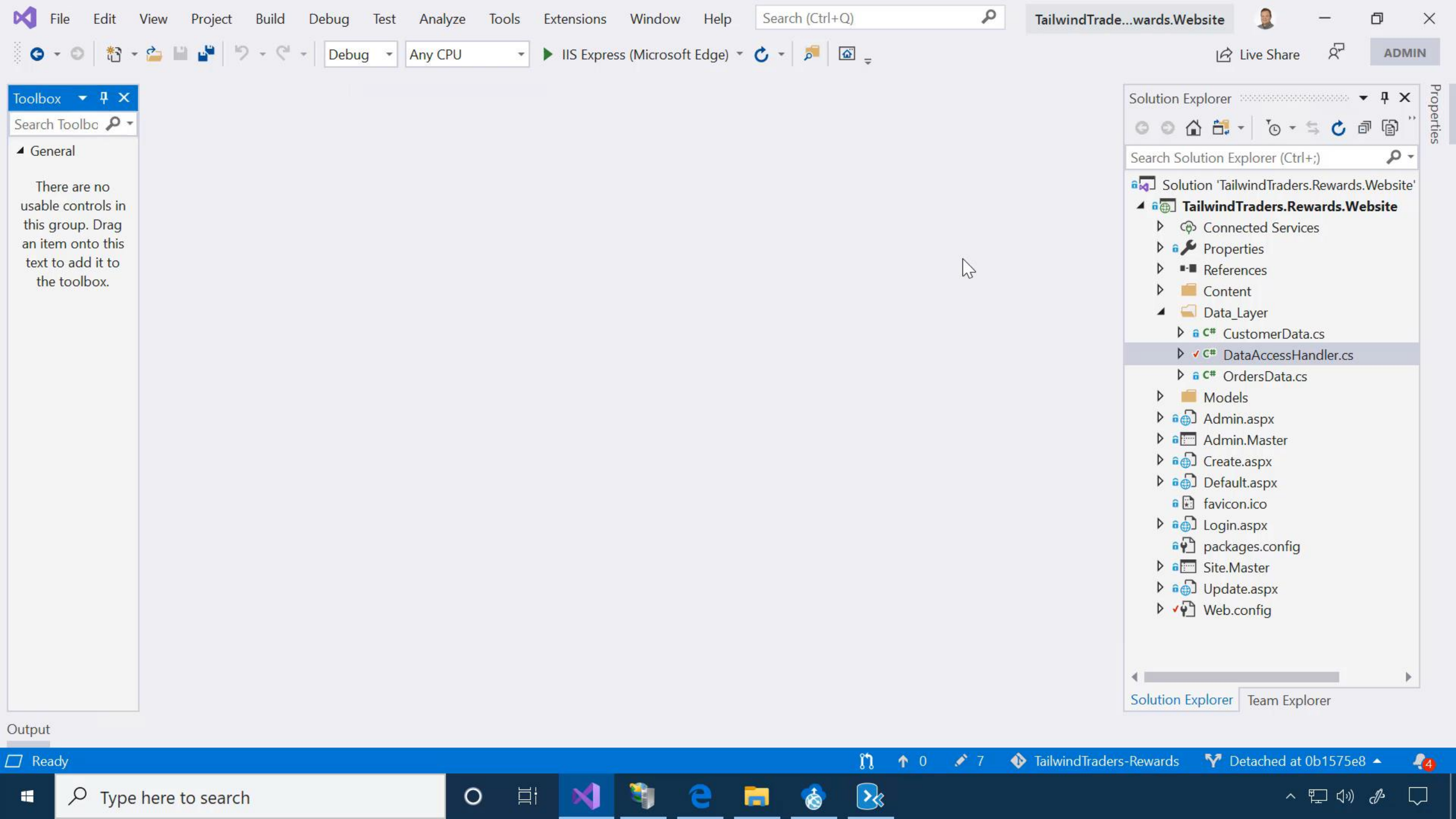
Useful links

Need help? Understand ownership cost... Learn about the available tools...



DEMO

Migrating the web app





Hosting your Web Applications using PaaS

We have options

What options do we have for managing our cloud applications?

The traditional tenants of hosting

Choosing the right balance of control and responsibility

Responsibility

On-prem

IaaS

Build from the ground up

PaaS

Some assembly required

SaaS

Move-in ready

Applications

Data

Runtime

Middleware

Operating system

Virtualization

Servers

Storage

Networking

Customer Microsoft

Azure App Service benefits



High-
productivity for
devs & ops



.NET, Node, Java, Docker,
PHP, Ruby, Python



Deploy containers on
Windows & Linux



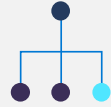
Staging &
deployment



Testing in
production



App gallery
marketplace



Fully-managed



Auto scale &
load balancing



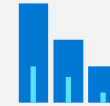
High availability
w/auto patching



Reduced
operations
costs



Backup &
recovery



Enterprise-grade



Global data
center footprint



Hybrid
support



Azure Active
Directory
integration



Secure &
compliance

Multiple deployment options



“Right-click publish” from Visual Studio

OK during development, don't abuse it.



Deployment center with various options

Azure repos, GitHub, OneDrive, Dropbox, FTP...



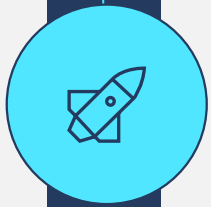
GitHub actions

Automated development directly from GitHub



Azure DevOps

Full pipelines with releases, gated deployment to slots etc



Deliver



Azure Boards



Azure Repos



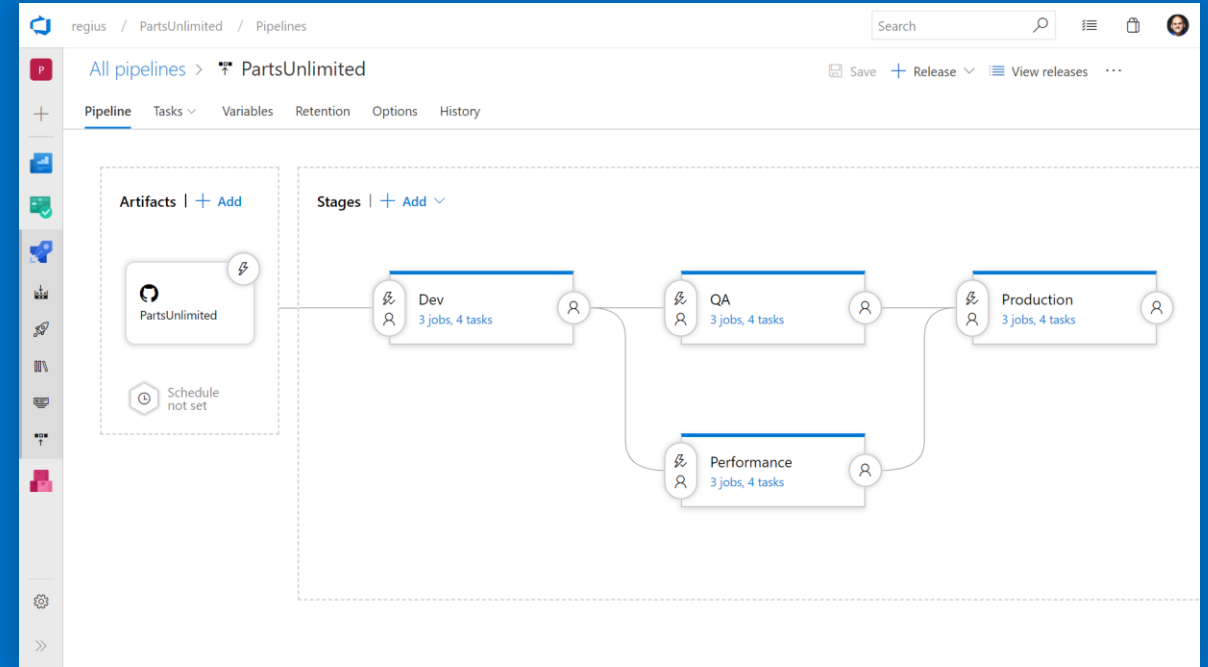
Azure Pipelines



Azure Artifacts



Azure Test Plans



Deploy to on-premises, ANY cloud or hybrid
from Linux, Mac and Windows

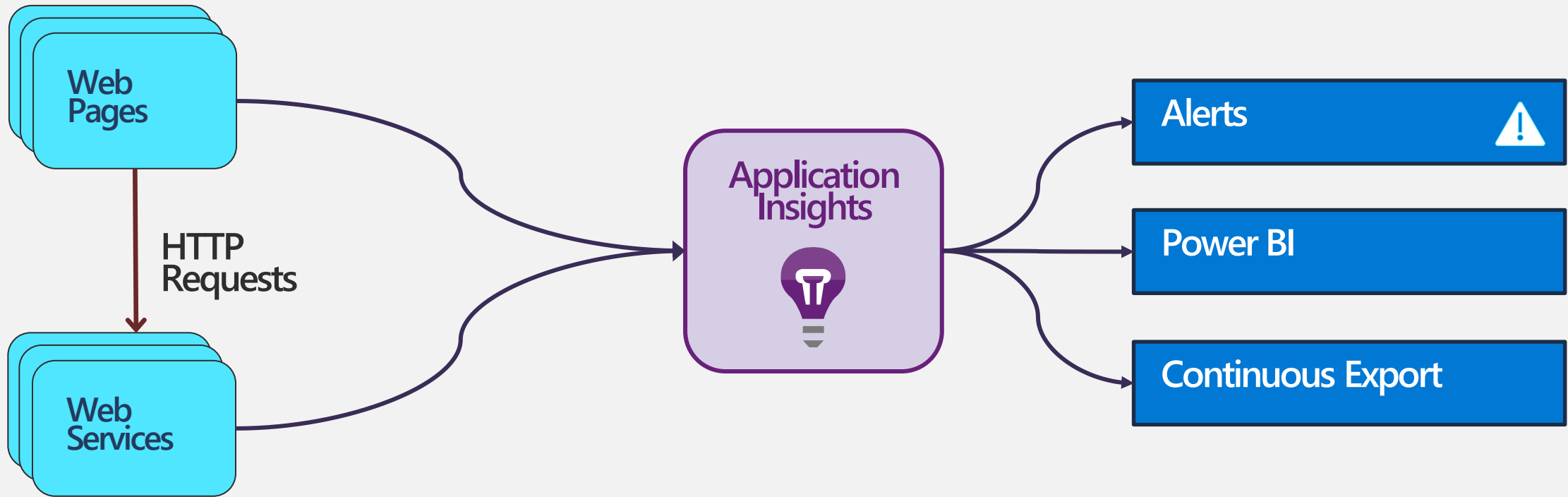
Staged environment releases

Deployment approvals including automated
options

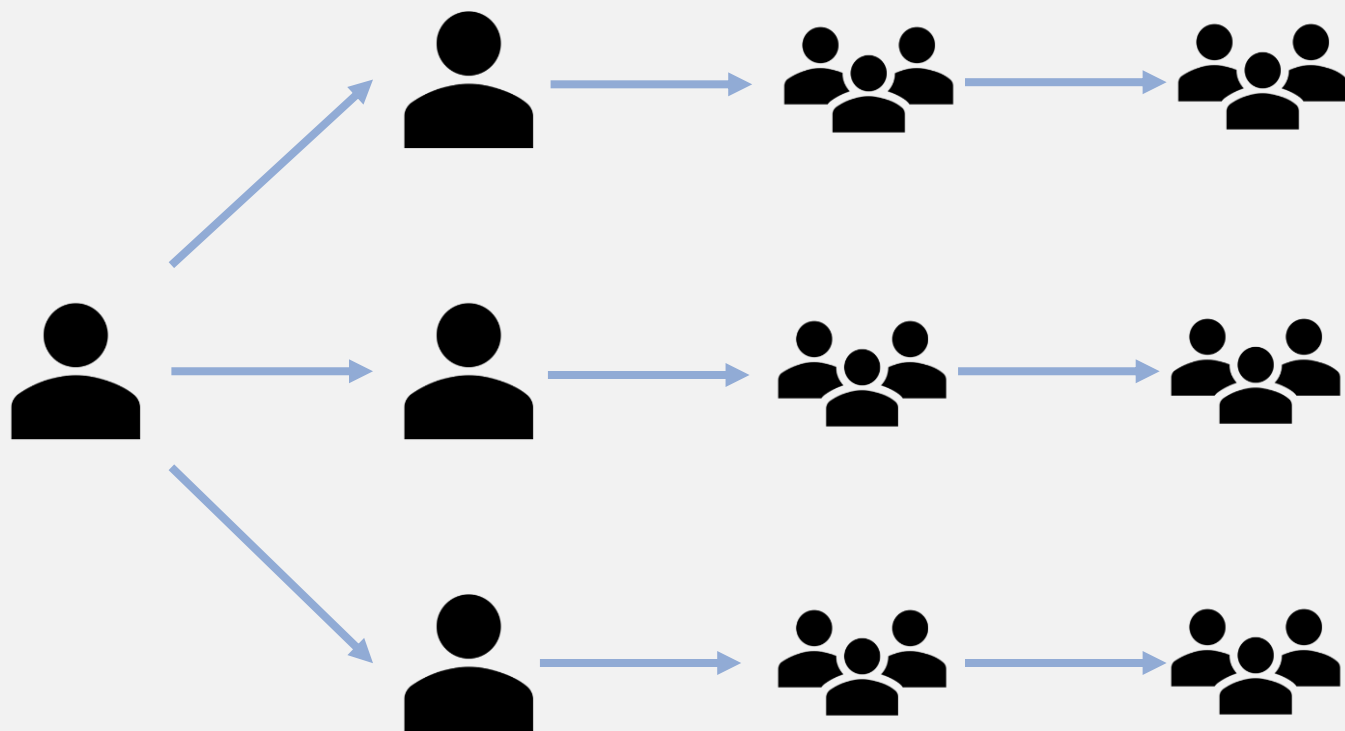
Native support for containers and Kubernetes



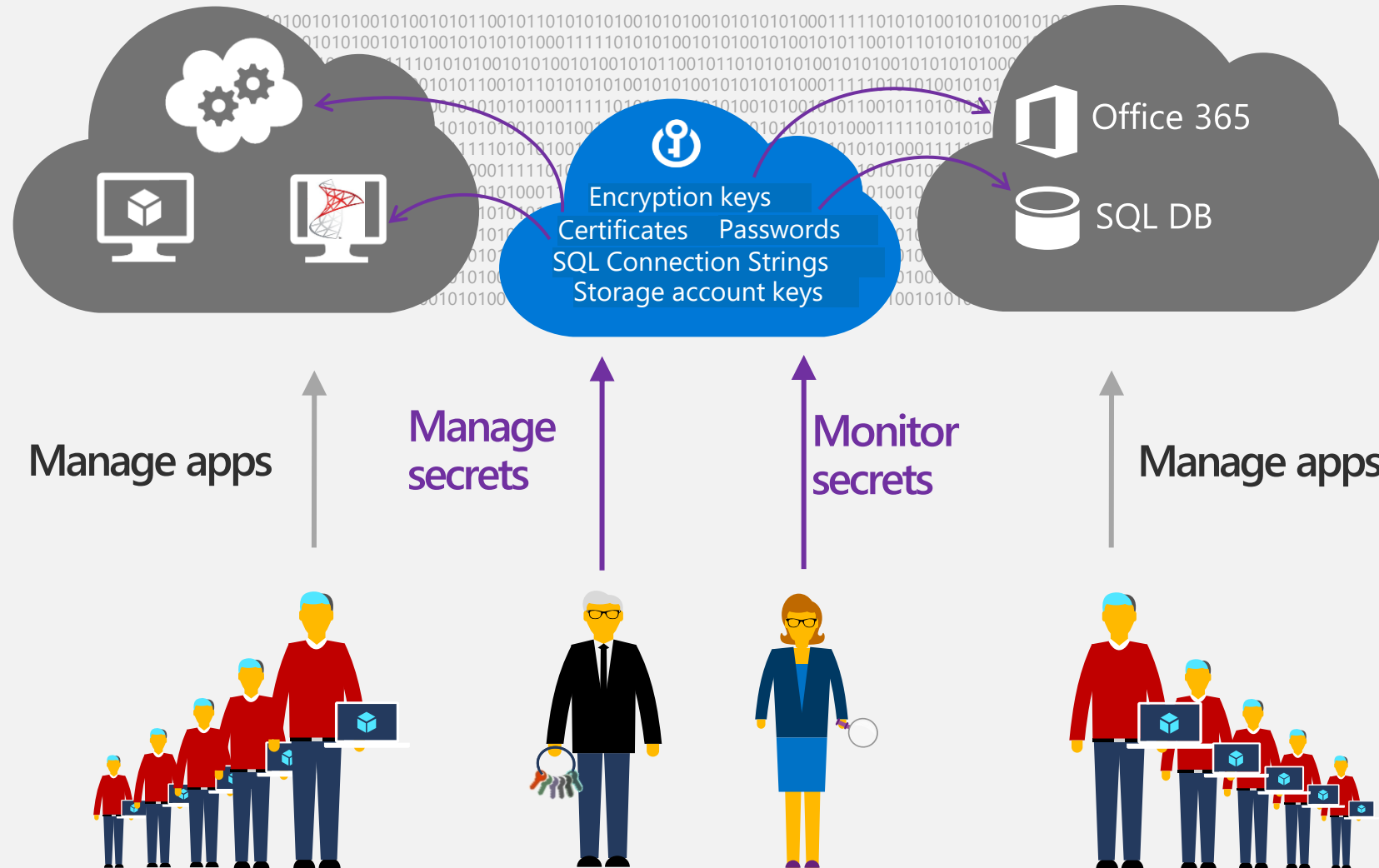
Application Insights



Basic Security Feature



Key Vault keeps you in control



DEMO

**Configuring the app
and Key Vault**

Scaling up and out



 tailwindrewardsonazure

App Service

 Search (Ctrl+/)

- Overview
- Activity log
- Access control (IAM)
- Tags
- Diagnose and solve problems
- Security
- Events (preview)
- Deployment
- Quickstart
- Deployment slots
- Deployment Center
- Settings

 Browse  Stop  Swap  Restart  Delete  Get publish profile  Reset publish profile

 .NET Framework 4.8 is coming to App Service starting in late July 2020 and will complete around late September 2020. Click the deployment.

Resource group (change)	URL
Rewards	https://tailwindrewardsonazure.azurewebsites.net/
Status	App Service Plan
Running	tailwindrewardsonazure-a0c0 (P1v2: 1)
Location	FTP/deployment username
West Europe	tailwindrewardsonazure\lbugnion
Subscription (change)	FTP hostname
ca-lbugnion-demo-test	ftp://waws-prod-am2-061.ft.azurewebsites.net/
	FTPS hostname
	ftps://waws-prod-am2-061.ft.azurewebsites.net/
Tags (change)	
migration : 1.0.6	





Contact info

Name Sophie
Last Name Stevenson
Email Address sophie@workingdata.com
Phone No 4251231234
Fax No 4259990000
Mobile No 4253214321

☐ Enroll in loyalty program

Client Info

Client's Acc No AC234148

[View another client](#)

Order history

ORDER N...	ORDER D...	ORDER T...	ITEM NA...	STATUS
192384529	8/11/2020...	Retail	Hammer	Placed
18983213	8/8/2020 ...	Heating	Stove	Placed
18676990	7/20/202...	Security	Plug-in In...	Placed
142184222	8/7/2020 ...	Sheds	Storage S...	placed

In review...



In review...



Moving your web application to App Service



Preparing application telemetry



Managing your settings and secrets



Scaling up and Scaling out

Call to action!

Explore overviews,
tutorials, samples, and more.

aka.ms/mod20

Microsoft App Service Migration Home Assess Download Readiness Checks Partners

Migrate to Azure App Service

Assess any app with an endpoint scan. Download the Migration Assistant and start your .NET and PHP app migration to Azure App Service.

Assess

Provide a public URL to assess to start your migration process.

Download

For internal .NET and PHP apps, download the Migration Assistant.

Start your assessment by entering a public facing endpoint to scan.

Assess your site by entering a public URL Assess

[Explore >](#) [Find Migration Partners](#) [Azure Migrate](#)



Using custom domains in Azure

Custom domain name

Azure App Service domain:

<https://myapservice.azurewebsites.net>

Custom domain:

<https://www.tailwindtraders.xyz>

In this session...



Basics of Domain Name Services (DNS)



Custom Domains in Azure Services



Adding TLS (Security Certificates) for https



Azure Traffic Manager



Azure Front Door



Domain Name Servers/Services (DNS)

TLD: Top Level Domain

- .com
- .org
- .net
- .xyz
- 100's more

A Records

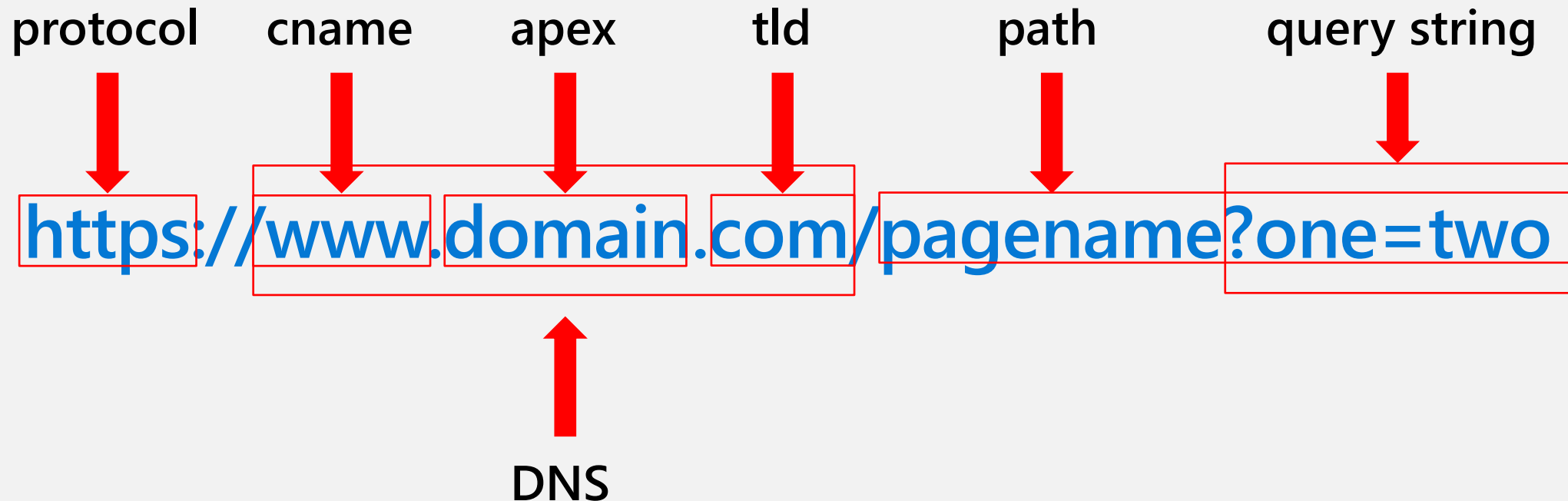
- Apex Record
 - A Record
 - The top of the highest part
- Root Domain
- Resolves to an IP
 - TailwindTraders.xyz
- CNAME Flattening
 - Experimental
 - Push www traffic to Root

CNAME Records

- Canonical Name Record
 - CNAME Record
 - Resolves to another alias
- Subdomains
 - www or api
 - us.north.api (multi level)
- www.tailwindtraders.xyz
 - > tailwindtraders.azurewebsites.net
- api.tailwindtraders.xyz
 - > twtfunctions.azurewebsites.net

What's in a URL

Uniform Resource Locator





Domain Name Verification

Why this is important?

You are proving that you “own the traffic” that comes into Azure on this domain.

You can not use a domain unless you can verify ownership

Verification Methods

- Validate via TXT record (Most Common, instant results)
- Other Validation methods

DEMO

Custom Domain with Azure App Service





TailwindTraders.xyz

Private dashboard

[+ New dashboard](#) [Refresh](#) [Full screen](#) [Edit](#) [Share](#) [Download](#) [Clone](#) [Assign tags](#) [Delete](#) [Feedback](#)Auto refresh : **Off**

Last updated

www-tailwindtraders

Web App

Running



functions-tailwindtr...

Function App

Running



storagetailwindtraders

Storage account



central-tailwindtrad...

Web App

Running



europe-tailwindtrad...

Web App

Running



app-trafficmanager-...

Traffic Manager profile



TailwindTraders.xyz



Resources

TailwindTraders.xyz

central-tailwindtraders	App Service	Central US
functions-tailwindtraders	Function App	Central US
TailwindTraders.xyz	DNS zone	Global
storagetailwindtraders	Storage account	
app-trafficmanager-xyz	Traffic Manager profile	Global
europe-tailwindtraders	App Service	West Europe
www-tailwindtraders	App Service	Central US
central-tailwindtraders	App Service plan	Central US
europe-tailwindtraders	App Service plan	West Europe
functionstailwindtraders	Storage account	Central US



Transport Layer Security (TLS) Certificates

TLS is an updated more secure Secure Socket Layer (SSL)

Certificate used to encrypt web traffic and enable https traffic

Binding a certificate to a custom domain options

- Bring your own certification
- Built certificate in Azure
- Store in Azure Key Vault for secure and central storage
- How to Update
- Avoid letting certificates expire

DEMO

Custom Domain Certificate on Azure App Service



DEMO

Custom Domain with Azure Functions



DEMO

Custom Domain with Azure Storage



Networking Strategies



Hardcoding to IP addresses will haunt you

App Service Custom Domains (+ Testing, Staging, Slots)

DNS with Azure Storage Custom Domains

CDN Custom Domains

Azure Function Custom Domains

Load Balancing Options



<https://aka.ms/mod40/loadbalancer>

Service	Global / Regional	Recommended Traffic
Azure Front Door	Global	HTTP(S)
Traffic Manager	Global	Non-HTTP(S)
Application Gateway	Regional	HTTP(S)
Azure Load Balancer	Global	Non-HTTP(S)



Azure Traffic Manager

DNS Based Load Balancing across multiple sources

Example:

<https://app.tailwindtraders.xyz>

- <https://app-region1.azurewebsites.net>
- <https://app-region2.azurewebsites.net>
- <https://app-region3.azurewebsites.net>
- Any HTTP address, not limited to Azure Properties

DEMO

Load Balancing

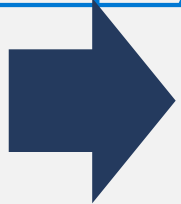


Azure Front Door

Routing based on path traffic

Example:

<https://app.tailwindtraders.xyz/images>



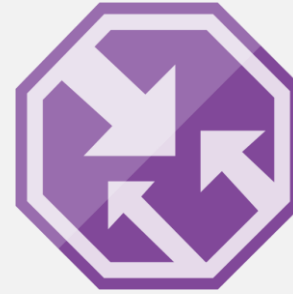
<https://image-cdn-storage.azureedge.net>

<https://app.tailwindtraders.xyz/scripts>



<https://scripts-cdn-storage.azureedge.net>

In review...



In review...



Basics of Domain Name Services (DNS)



Custom Domains in Azure Services



Adding TLS (Security Certificates) for https



Azure Traffic Manager

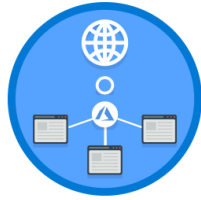


Azure Front Door

Call to action!

Explore overviews,
tutorials, samples, and more.

aka.ms/mod40



Load balance your web service traffic with Application Gateway

1 hr 32 min • Module • 7 Units

★★★★☆ 4.6 (2,069)

Rate it

Beginner

Solutions Architect

Administrator

Azure

Virtual Network

Application Gateway

Improve application resilience by distributing load across multiple servers and use path-based routing to direct web traffic.

Learning objectives

In this module, you will:

- Identify the load balancing capabilities of Application Gateway
- Create an Application Gateway and configure load balancing
- Configure an Application Gateway to use URL path-based routing

Start >

Bookmark

Add to collection

800 XP

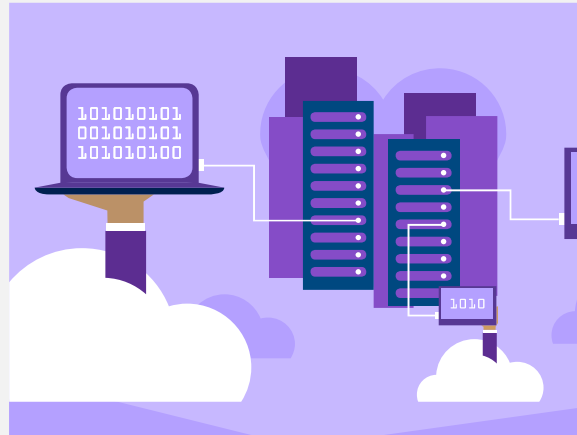


Monitoring the Health and Performance of your Web Application

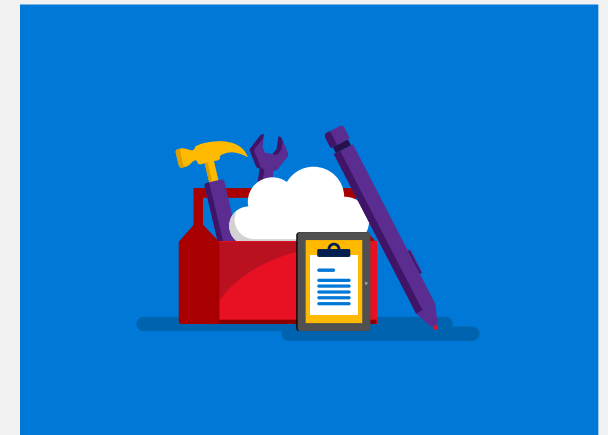
Working with production apps can be hard



Frustrations with
getting exceptions /
logs / data



Issues caused by
config, data, and scale



Acquiring and
configuring the
tools

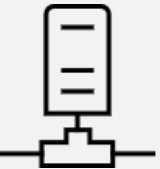
Monitoring the whole landscape



Infrastructure (on-premise and in the cloud)



Applications



Network

In this session...



A single solution with Azure Monitor



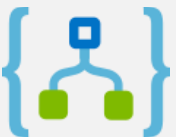
See inside your production apps with Application Insights



Query logs and traces with Log Analytics



Debug apps in production with Snapshot debugging



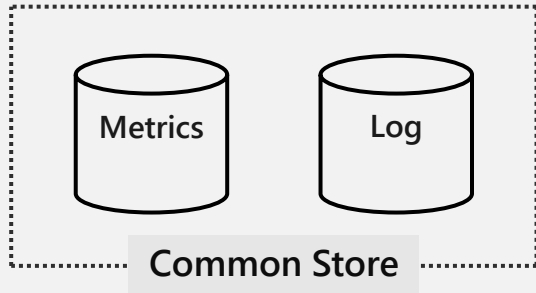
Act with notifications and Logic Apps



Azure Monitor and Application Insights

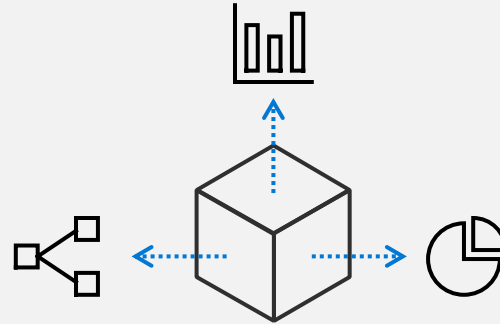
Azure Monitor

Full observability for your infrastructure, app and network



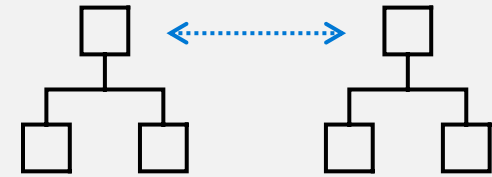
Unified Monitoring

A common platform for all metrics, logs and other monitoring telemetry



Data Driven Insights

Advanced diagnostics and analytics powered by machine learning capabilities



Workflow Integrations

Rich ecosystem of popular DevOps, issue management, SIEM, and ITSM tools

Unified monitoring

Application

Operating System

Azure Resources

Azure Subscription

Azure Tenant

Custom Sources



Insights



Application



Container



VM



Monitoring
Solutions

Visualize



Dashboards



Views



Power BI



Workbooks

Analyze



Metrics Explorer



Log Analytics

Respond



Alerts



Autoscale

Integrate



Event Hubs



Logic Apps



Ingest &
Export APIs

Signals and sources in Azure

Signals



Metrics



Logs



Alerts

Sources

Application	User telemetry
	Application logs

Guest OS (‘user space’)	Linux syslog
	Windows Perf Counters

Azure Resources	Virtual Machines
	Storage Accounts
	Network Security Groups

Azure Subscriptions	Resource Manager
	Service Health
	Security Center

Azure Tenants	Azure Active Directory
---------------	------------------------



Visibility – Get the big picture



Insights – Find & fix problems



Optimization – Build, measure, learn

Azure Application Insights



Source of telemetry



- 1 Outside-in monitoring
- 2 Observed user behavior
- 3 Developer traces and events
- 4 Observed application behavior
- 5 Infrastructure performance

Correlated tracing across servers



Dependency Properties

visitorswebdiag

+ New Work Item View Work Items

HTTP

Event Time: 5/1/2017, 8:49:49 AM < 1 of 28 >

Dependency Properties

Event time	5/1/2017, 8:49:49 AM	...
Dependency type	HTTP	...
Successful call	false	...
Result code	500	...
Dependency duration	29.35 s	...
Dependency name	http://visitorscrmdia.azurawebsites.net:80	...

Custom Data

_MS.ProcessedByMet...	(Name:'Dependencies', Ver:'1.0')	...
-----------------------	----------------------------------	-----

Path

http://visitorscrmdia.azurawebsites.net:80/api/crmdata/1

Related Items

Request in which this dependency call was made	1	●
Failed or slow calls to remote dependencies	1	●
All available telemetry for this operation	1	●
All available telemetry for this root operation	7	●
All available telemetry 5 minutes before and after this event	73	●

Search

4/30/2017 8:49 PM to 5/1/2017 8:49 PM (1 hour granularity) - visitorswebdiag

Time range Filters Refresh Reset Analytics More

Filtered on Trace x Request x Page View x Custom Event x Exception x
Dependency x Availability x With D0/986edx2Q= x

7 total results between 4/30/2017 8:49:49 PM and 5/1/2017 8:49:49 PM (1 day)

5/1/2017, 8:50:18 AM - EXCEPTION
The provider did not return a ProviderManifestToken string.
Exception type: System.ComponentModel.Win32Exception
Failed method: MyCompany.Visitors.CRMServiceFx.Controllers.CRMDataController+<GetCRMDatA...
Problem Id: System.Data.Entity.Core.ProviderIncompatibleException at System.Data.Entity.Core....

5/1/2017, 8:50:18 AM - EXCEPTION
An error occurred accessing the database. This usually means that the connection to the database failed. Check that the connection string is correct and that the appropriate DbContext constructor is being used to specify it or find it in...
Exception type: System.ComponentModel.Win32Exception
Failed method: MyCompany.Visitors.CRMServiceFx.Controllers.CRMDataController+<GetCRMDatA...
Problem Id: System.ComponentModel.Win32Exception at MyCompany.Visitors.CRMServiceFx.Contr...

5/1/2017, 8:50:18 AM - EXCEPTION
An error occurred accessing the database. This usually means that the connection to the database failed. Check that the connection string is correct and that the appropriate DbContext constructor is being used to specify it or find it in...
Exception type: System.ComponentModel.Win32Exception
Failed method: MyCompany.Visitors.CRMServiceFx.Controllers.CRMDataController+<GetCRMDatA...
Problem Id: System.Data.Entity.Core.ProviderIncompatibleException at System.Data.Entity.Utili...

5/1/2017, 8:49:49 AM - REQUEST
GET crmdata [id]
Request URL: http://visitorscrmdia.azurawebsites.net/api/crmdata/1 Response code: 500
Server response time: 29.3 s Request URL path: /api/crmdata/1

5/1/2017, 8:49:49 AM - EXCEPTION
A network-related or instance-specific error occurred while establishing a connection to SQL Server. The server was not found or was not accessible. Verify that the instance name is correct and that SQL Server is configured to allow remote...
Exception type: System.ComponentModel.Win32Exception
Failed method: MyCompany.Visitors.CRMServiceFx.Controllers.CRMDataController+<GetCRMDatA...
Problem Id: System.Data.SqlClient.SqlException at System.Data.SqlClient.SqlInternalConnection...

5/1/2017, 8:49:49 AM - DEPENDENCY
visitorscrmdia.azurawebsites.net
Dependency duration: 29.35 s Successful call: false
Dependency name: http://visitorscrmdia.azurawebsites.net:80

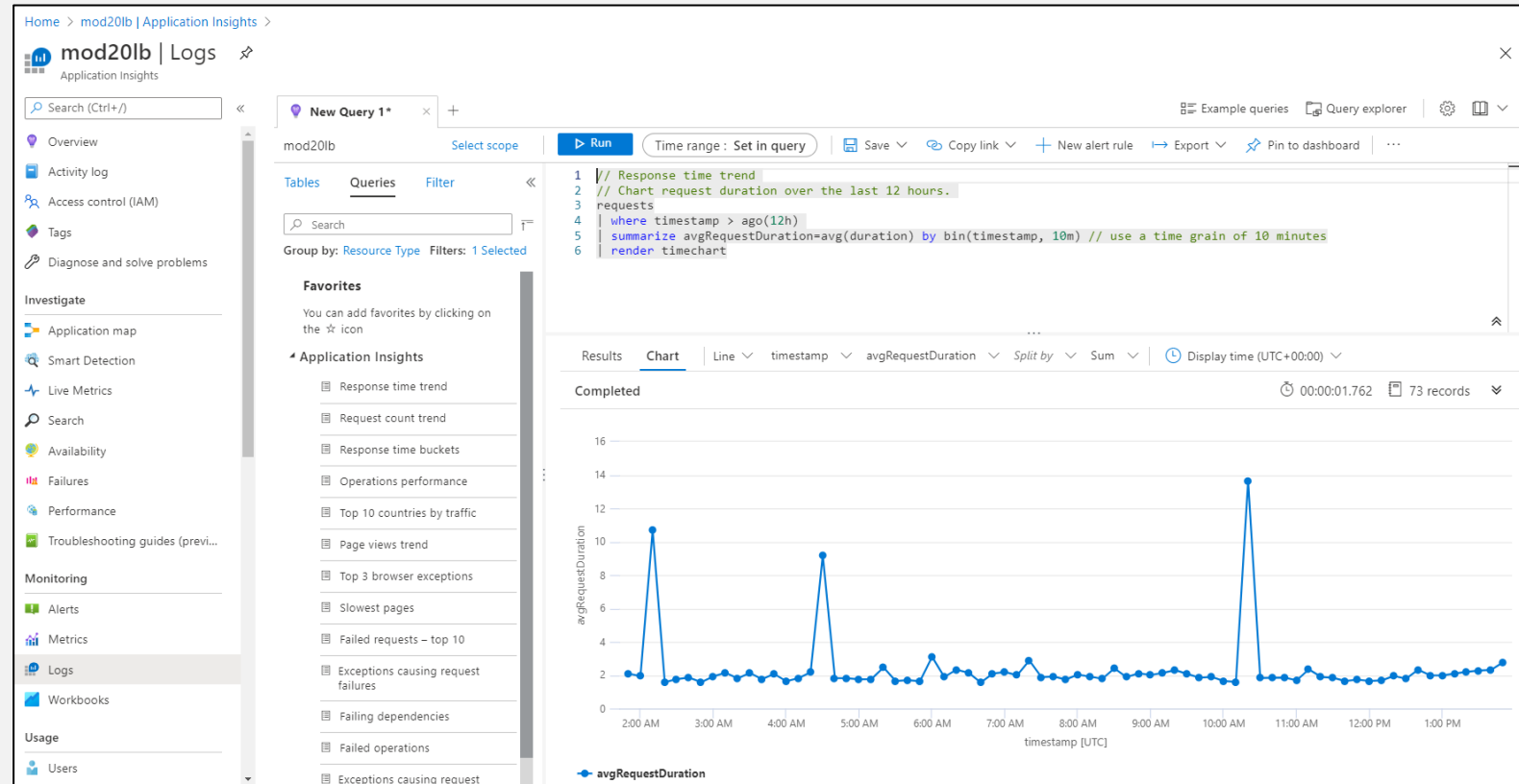
5/1/2017, 8:49:49 AM - REQUEST
GET /noauth/api/crm/1
Request URL: http://visitorswebdiag.azurawebsites.net/noauth/api/crm/1 Response code: 200
Server response time: 29.35 s Request URL path: /noauth/api/crm/1

Azure Monitor Logs



Log Analytics

- ➔ Log Analytics advanced query experience now in Azure Portal
- ➔ RBAC per type
- ➔ Run analytics queries for investigations, statistics, and root cause + trend analysis
- ➔ Utilize ML algorithms for clustering and anomaly detection



Using Analytics to Dig Deeper



Home Page x New Query 1 * x New Query 2 * x +

exceptions
| evaluate autocluster()

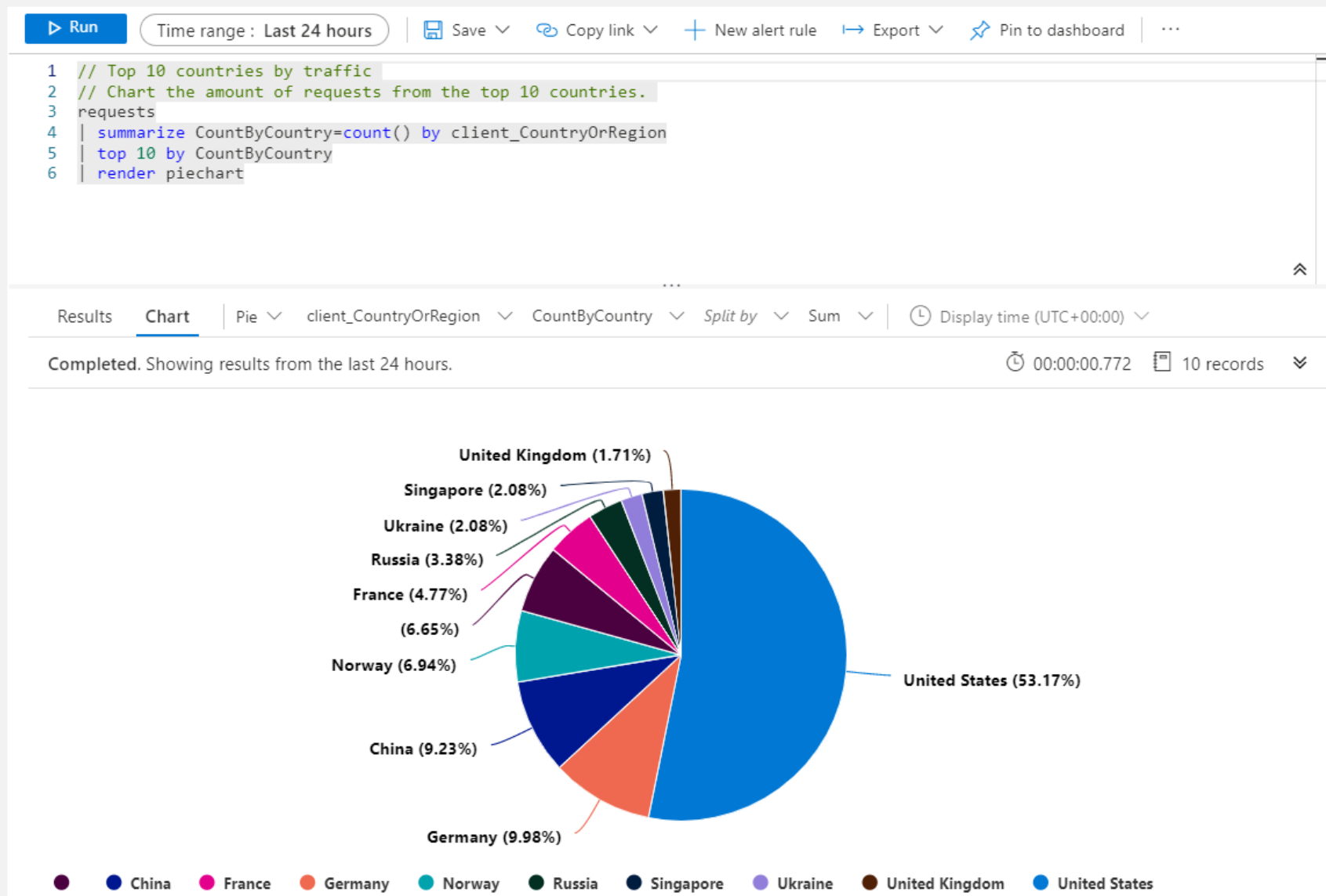
Completed. Showing results from the last 24 hours.

TABLE CHART

Drag a column header and drop it here to group by that column

Count (of pattern)	Percent (%)	problemId	cloud_RoleName	ha
686	24.4912531239	System.ComponentModel.Win32Exception at MyCompany.Visitors.CRMSvcFx.Controllers.CRMDDataController+<GetCRMDDataAsy...	visitorscrmdiag	
685	24.4555515887	System.Data.SqlClient.SqlException at System.Data.SqlClient.SqlInternalConnection.OnError:39	visitorscrmdiag	
683	24.3841485184	System.Data.Entity.Core.ProviderIncompatibleException at System.Data.Entity.Core.Common.DbProviderServices.GetPr...	visitorscrmdiag	
683	24.3841485184	System.Data.Entity.Core.ProviderIncompatibleException at System.Data.Entity.Utilities.DbProviderServicesExtensio...	visitorscrmdiag	
2,740	97.8222063549	*	visitorscrmdiag	
2,801	100	*	*	

Azure Monitor Logs





DEMO

Exploring Azure Monitor and Application Insights

Overview

Activity log

Access control (IAM)

Tags

Events

Settings

Quickstart

Deployments

Policies

Properties

Locks

Export template

Cost Management

+

 Add

≡

 Edit columns

🗑

 Delete resource group

🔄

 Refresh

➡

 Move

⬇

 Export to CSV

⋮

Essentials

Type == all

Location == all

+ Add filter

Showing 1 to 11 of 11 records.

☐ Show hidden types

No grouping

List view

☐ Name ↑↓	Type ↑↓	Location ↑↓
☐  lbrewardssql	SQL server	West Europe
☐  NIC-3r3km3qe4qrmips3xi7u4xmp	Network interface	West Europe
☐  Rewards-Key-Vault	Key vault	West Europe
☐  rewardsdb (lbrewardssql/rewardsdb)	SQL database	West Europe
☐  RewardsDms	Azure Database Migration ...	West Europe
☐  RewardsMig (RewardsDms/RewardsMig)	Azure Database Migration ...	West Europe
☐  RewardsVNet	Virtual network	West Europe

 Search (Ctrl+ /)



-  Browse
-  Stop
-  Swap
-  Restart
-  Delete
-  Get publish profile
-  Reset publish profile
- ...

-  Overview
-  Activity log
-  Access control (IAM)
-  Tags
-  Diagnose and solve problems
-  Security
-  Events (preview)
- Deployment
-  Quickstart
-  Deployment slots
-  Deployment Center
- Settings
-  Configuration

 .NET Framework 4.8 is coming to App Service starting in late July 2020 and will complete around late September 2020. Click [to learn more and see progress on the deployment.](#) 

Resource group (change)	URL
Rewards	https://tailwindrewardsonazure.azurewebsites.net
Status	App Service Plan
Running	tailwindrewardsonazure-a0c0 (P1v2: 2)
Location	FTP/deployment username
West Europe	tailwindrewardsonazure\lbugnion
Subscription (change)	FTP hostname
ca-lbugnion-demo-test	ftp://waws-prod-am2-061.ftp.azurewebsites.windows...
	FTPS hostname
Tags (change)	ftps://waws-prod-am2-061.ftp.azurewebsites.window...
migration : 1.0.6	





Snapshot debugging

Snapshot debugging

Debug in
Production



Snapshot debugging

Debug in
Production

Distributed
App
Support



Snapshot debugging

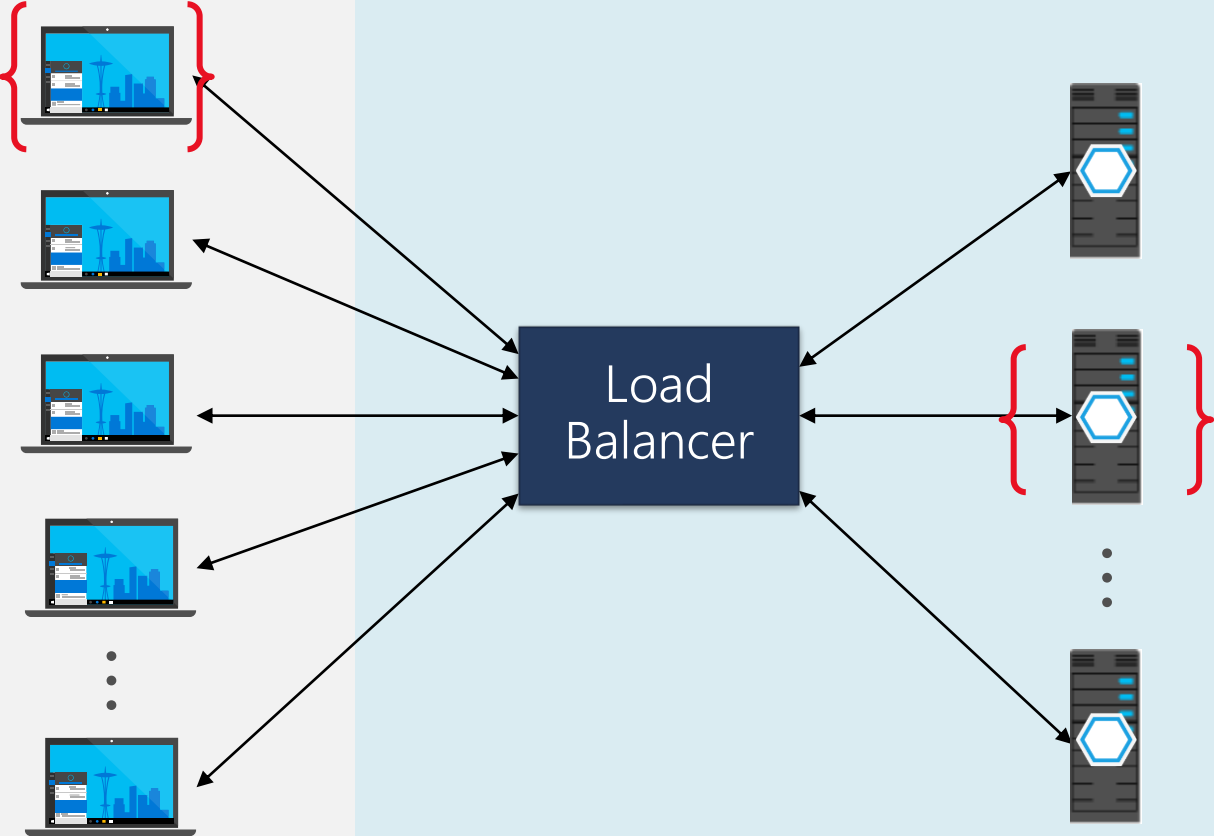
Debug in
Production

Distributed
App
Support

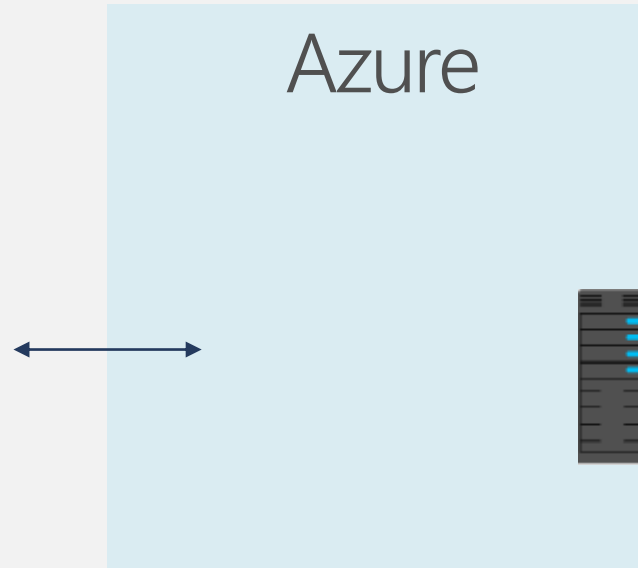
Secure
Access by
Role



Azure



Snapshot debugging





DEMO

Going back in time with Snapshots

 Free Shipping • For \$75 or more orders



start smart shopping
By uploading a photo

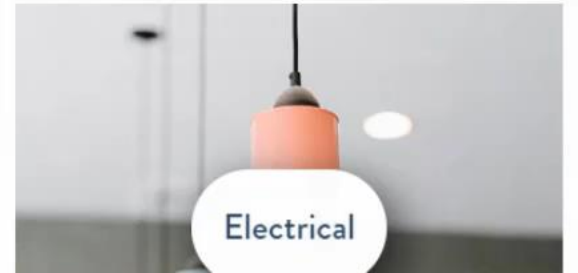
RECOMMENDED



Plumbing



Electrical



Notifications and complex actions

Acting on errors



Logs are great but we need notifications!!

Email, SMS

Azure app push notifications

Complex actions with logic apps, functions...

Monitoring availability

tailwindrewards

Availability

Application Insights

Search (Ctrl+ /)

<<

+ Add test

🔄 Refresh

😊 Feedback

Overview

Activity log

Access control (IAM)

Tags

Diagnose and solve problems

Investigate

Application map

Smart Detection

Live Metrics

Search

Availability

Failures

Performance

Troubleshooting guides (previ...

Monitoring

Alerts

Metrics

Logs

Workbooks

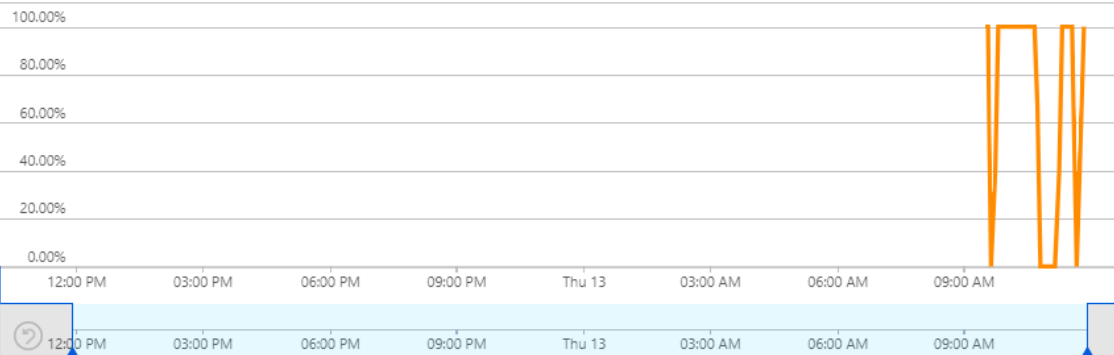
Usage

Users

Local Time: Last 24 hours

Availability ⓘ

LineScatter Plot



Select availability test

Search to filter items...

AVAILABILITY TEST	20 MIN	AVAILABILITY	DURATION (AVG)
Overall	47.37%	64.83%	1.08 sec
▼ ⚠ TailwindRewardsAvailability	47.37%	64.83%	1.08 sec
⚠ Central US	33.33%	65.52%	522 ms
⚠ East US	50.00%	62.07%	445 ms
⚠ North Central US	50.00%	65.52%	497 ms
⚠ South Central US	50.00%	65.52%	3.22 sec
⚠ West US	50.00%	65.52%	717 ms

Overall

Availability results

	COUNT
Successful	94
Failed	51

Drill into...

94 Successful51 Failed

Handling alerts

[Home](#) >

 **tailwindrewards** | Alerts 

Application Insights



[+ New alert rule](#) [Manage alert rules](#) [Manage actions](#) [View classic alerts](#) [Refresh](#) [Provide feedback](#)

 Overview

 Activity log

 Access control (IAM)

 Tags

 Diagnose and solve problems

Investigate

 Application map

 Smart Detection

 Live Metrics

 Search

 Availability

 Failures

 Performance

 Troubleshooting guides (previ...

Don't see a subscription? [Open Directory + Subscription settings](#)

Subscription * 

ca-lbagnion-demo-test 

Resource group 

TailwindRewards 

Resource 

tailwindrewards 

Time range 

Past 24 hours 

[Selected subscriptions](#) > [TailwindRewards](#) > tailwindrewards

Total alerts

3

Since 8/12/2020, 12:12:43 PM

Smart groups (preview) 

2 

33.33% Reduction

Total alert rules

1

Enabled 1

Action rules (preview) 

0

Enabled 0

[Learn more About alerts](#) 

Severity	Total alerts	New	Acknowledged	Closed
 Sev 0	0	0	0	0
 Sev 1	0	0	0	0
 Sev 2	0	0	0	0
 Sev 3	0	0	0	0
 Sev 4	3	3	0	0

Sending notifications

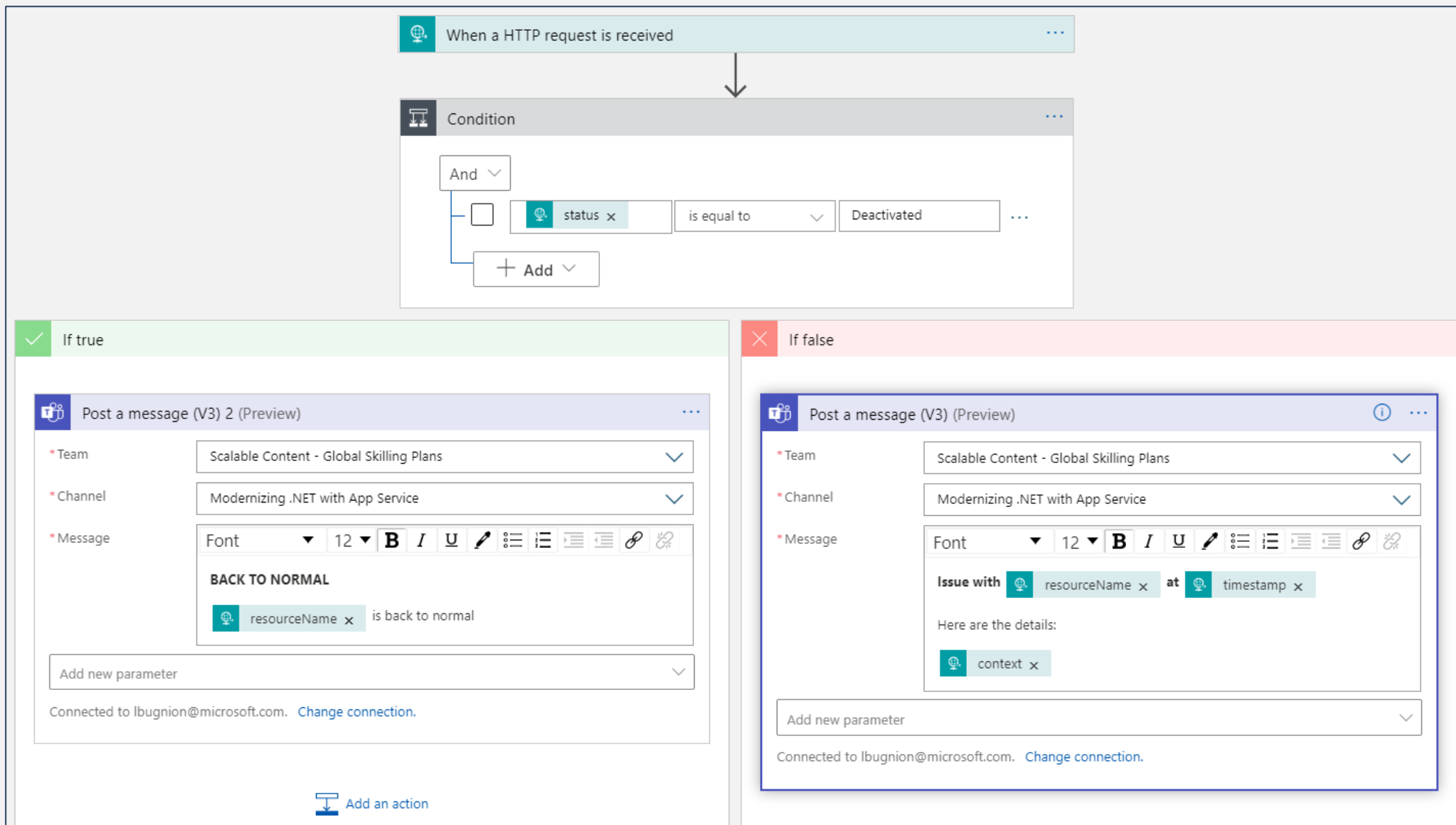
Notifications

Notification type	Name	Selected	
Email Azure Resource Manager Role	Monitoring Contributor	Monitoring Contributor	 
Email Azure Resource Manager Role	Monitoring Reader	Monitoring Reader	 
Email/SMS message/Push/Voice	Alert the owner	Email, Push ⓘ	 

Actions

Action type	Name	Selected	
Logic App	Complex action	TailwindRewardsAlerts	 

Complex actions





DEMO

Notifications and complex actions



Contact info

Name John
Last Name Doe
Email Address john@workingdata.co.uk
Phone No 4251231234
Fax No 4259990000
Mobile No 4253214321

☐ Enroll in loyalty program

Client Info

Client's Acc No AC234323

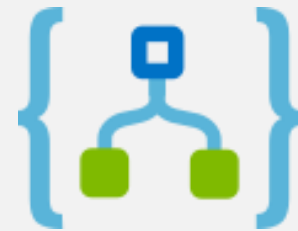
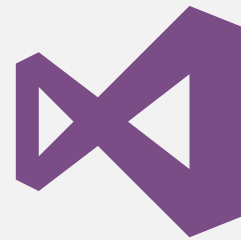
[View another client](#)



Order history

ORDER N...	ORDER D...	ORDER T...	ITEM NA...	STATUS	TOTAL
192384529	8/11/202...	Retail	Hammer	Placed	100.00
18903849	8/13/202...	Web	Wrench	--	150.00
18983945	8/12/202...	Phone	Cordless ...	--	225.00
18983990	8/11/202...	Web	Hammer	--	100.00

In review...



In review...



A single solution with Azure Monitor



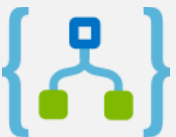
See inside your production apps with Application Insights



Query logs and traces with Log Analytics



Debug apps in production with Snapshot debugging



Act with notifications and Logic Apps